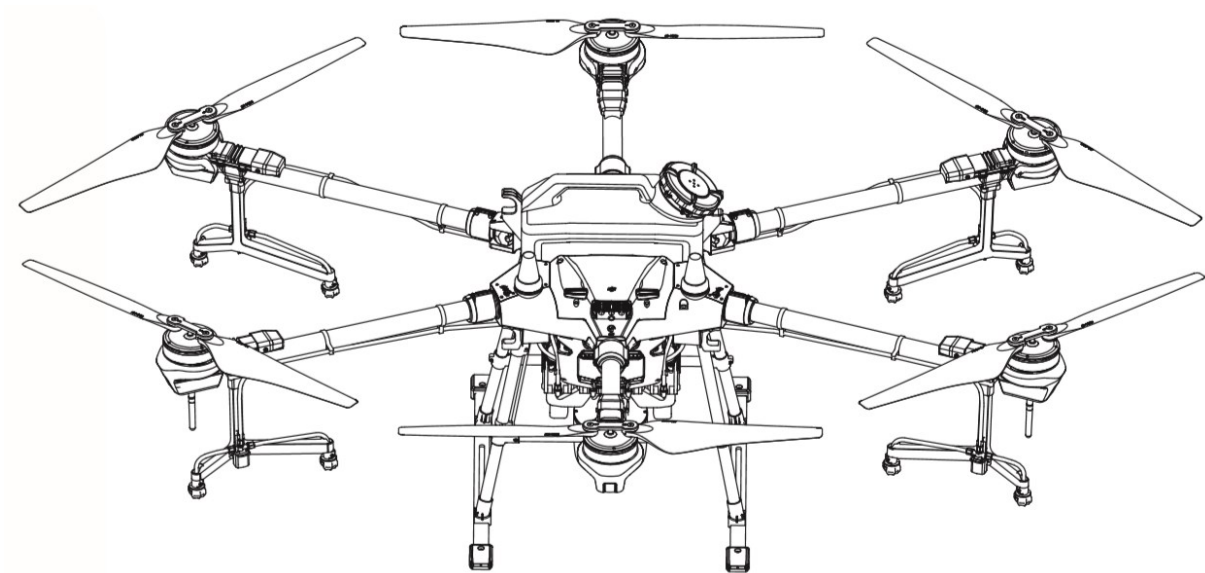


For DJI Authorized Dealers Use Only

T20 Repair Guide



新疆农业售后外发

For DJI Authorized Dealers Use Only

History				
Version	Content	Written On	Written By	Approved Nu
V1.0.0	First Edition	2020.10.21	GKAS	

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大疆农业售后外发文档 (代理商专用)

Purpose

- The purpose of this document is to show DJI authorized dealers how to repair the T20. This repair guide is only for dealers who have signed the confidentiality agreement and have a product repair qualification. This document cannot be distributed to other third parties or be disclosed to the public. If you find any mistakes or have any suggestions, please contact your regional manager.
- To find the related repair information and policies please visit: <https://dealersystem.dji.com/>

Repair Notices

- Repair and calibration should be carried out by qualified technical staff.
- Please ensure that all operations are in an anti-static workshop.
- Please ensure that no soldering tin, screws or debris are inside the product and its components.
- Please ensure that all modules, screws, and conductive foams are mounted in place and that glue is applied properly.
- Please ensure that all the cables are connected correctly.

Anti-Static Protection

- Disassembly or assembly of the external shell must be done in an anti-static workshop.
- The electronic components are electrical static sensitive devices. Electrostatic discharge (ESD) is the main cause of damage to electronic components.
- Ensure that anti-static measures are taken when transporting the electrical static sensitive devices.
- Do not operate in a strong static or magnetic field.

Maintenance

- Keep the drone and components away from children.
- Do not let the component come into contact with liquid.
- Do not power on the drone immediately after it has fallen into water.
- Do not store the camera in a humid area.

Battery

- Do not allow the battery to come into contact with any kind of liquid. Do not leave the battery out in the rain or make it wet.
- Do not use swollen, leaking, or damaged batteries.
- Do not install or remove the battery from the aircraft when it is turned on.
- The battery needs to be used at temperatures from -10°C to 40°C.

-
- Do not use the battery in strong electrostatic or electromagnetic environments.
 - Do not disassemble or pierce the battery in any way.
 - Do not place wires or metal objects on the battery, or the battery may short-circuit.
 - Do not drop or hit batteries. Do not place heavy objects on the battery.
 - Electrolytes in the battery are highly corrosive. If they leak, please stay away from them.
 - Keep batteries out of the reach of children.
 - Do not leave the battery near heat sources.
 - Keep the battery dry.
 - Keep batteries away from the metal objects, such as glasses, watches, jewelry, or hairpins.
 - Do not transport a damaged battery or a battery with a power level higher than 30%.
 - Do not store batteries in environments with a temperature higher than 45°C or lower than 0°C.
 - Do not place batteries in regular trash containers.

大疆农业售后外发文档 (代理商专用)

1 Product Information

FLIGHT PARAMETERS

Total Weight (Excluding battery)	21.1 kg
Standard Takeoff Weight	42.6 kg
Max Takeoff Weight	47.5 kg (at sea level)
Max Thrust-Weight Ratio	1.70 (with a takeoff weight of 47.5 kg)
Hovering Accuracy (With a strong GNSS signal)	D-RTK enabled: Horizontal: ± 10 cm, Vertical: ± 10 cm D-RTK disabled: Horizontal: ± 0.6 m, Vertical: ± 0.3 m (Radar module enabled: ± 0.1 m)
RTK/ GNSS Operating Frequency	RTK: GPS L1/L2, GLONASS F1/F2, BeiDou B1/B2, Galileo E1/E5 ^[1] GNSS: GPS L1, GLONASS F1, Galileo E1 ^[1]
Max Power Consumption	8300 W
Hovering Power Consumption	6200 W (with a takeoff weight of 47.5 kg)
Hovering Time ^[2]	15 min (takeoff weight of 27.5 kg with a 18000 mAh battery) 10 min (takeoff weight of 42.6 kg with a 18000 mAh battery)
Max Tilt Angle	15°
Max Operating Speed	7 m/s
Max Flying Speed	10 m/s (with a strong GNSS signal)
Max Wind Resistance	8 m/s
Max Service Ceiling Above Sea Level	2000 m
Recommended Operating Temperature	0° to 40°C

AIRFRAME

Max Diagonal Wheelbase	1883 mm
Dimensions	2509×2213×732 mm (Arms and propellers unfolded) 1795×1510×732 mm (Arms unfolded and propellers folded) 1100×570×732 mm (Arms and propellers folded)

PROPULSION SYSTEM - MOTOR

Stator Size	100×15 mm
Weight	666 g

PROPULSION SYSTEM

KV	75 rpm/v
Max Thrust	13.5 kg/rotor
Max Power	2400 W/rotor
Weight (Single propeller)	90 g

PROPULSION SYSTEM - PROPELLERS

Diameter × Pitch	33×9 in
------------------	---------

PROPULSION SYSTEM - ESC

Max Working Current (Continuous)	40 A
Max Working Voltage	58.8 V(14S LiPo)

FPV CAMERA

FOV	Horizontal: 98°, Vertical: 78°
Resolution	1280×960 30fps

FPV SPOTLIGHT

Max Brightness	12 lux at 5 m of direct light
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SPRAYING SYSTEM

Spray Tank Volume	Rated: 15.1 L, Full: 20 L
Operating Payload	Rated: 15.1 kg, Full: 20 kg



SPRAYING SYSTEM - NOZZLE

Nozzle Model	SX11001VS (standard) SX110015VS (optional) XR11002VS(optional)
Quantity	8
Max Spray Rate	SX11001VS : 3.6 L/min SX110015VS : 4.8L/min XR11002VS : 6 L/min
Droplet Size	SX11001VS : 130-250 µm SX110015VS : 170-265 µm XR11002VS : 190-300 µm (related to the actual working environment, spraying flow, etc.)
Spray Width	4-7 m (8 nozzles, at a height of 1.5 - 3 m above crops)

SPRAYING SYSTEM - FLOW METER

Measurement Range	0.25-20 L/min
Error	< ±2%
Measurable Liquid	Conductivity > 50 µS/cm (Liquids such as water or pesticides that contain water)



HIGH PRECISION RADAR MODULE

Model	RD2428R
Operating Frequency	CE (Europe)/(United States): 24.00 GHz-24.25 GHz MIC (Japan)/KCC (Korea): 24.05 GHz-24.25 GHz
Power Consumption	18W
EIRP	MIC/KCC/CE /FCC: <20 dBm
Altitude Detection & Terrain Follow ^[3]	Altitude detection range: 1-30 m Stabilization working range: 1.5-15 m Maximum slope of hill mode: 35°
Obstacle Avoidance System ^[3]	Altitude detection range: 1.5-30 m FOV: Horizontal: 360°, Vertical: ±15° Condition: Relative altitude higher than 1.5 m, operation speed lower than 7 m/s Safe distance: 2.5m (distance between the propeller tip and the obstacle after the aircraft stops) Obstacle-avoiding direction: Omnidirectional obstacle sensing operates horizontally and covers 360°
IP Rating	IP67

T20 Intelligent Flight Battery

Model	AB3-18000mAh-51.8V
Weight	6.4 kg (approximately)
Discharge Rate	9 C
IP Rating	IP54
Capacity	18000 mAh
Voltage	51.8 V

REMOTE CONTROLLER

Model	RM500-AG
OcuSync 2.0 Operating Frequency ^[4]	2.4000-2.4835 GHz 5.725-5.850 GHz
OcuSync 2.0 Effective Transmission Distance(Unobstructed, free of interference)	SRRC/MIC/KCC/CE: 3 km NCC/FCC: 5km (Operation height: 2.5 m)
OcuSync 2.0 Transmission Power (EIRP)	2.4 GHz SRRC/CE/MIC/KCC: 18.5 dBm, FCC: 25.5 dBm; 5.8 GHz SRRC/FCC: 25.5 dBm
WIFI Protocol	Wi-Fi Direct, Wireless Display, 802.11 a/g/n/ac, and Wi-Fi with 2×2 MIMO is supported
Wi-Fi Operation Frequency Range ^[4]	2.4000-2.4835 GHz 5.150-5.250 GHz 5.725-5.850 GHz
WIFI Transmission Power (EIRP)	2.4 GHz SRRC/CE: 18.5 dBm; FCC/MIC/KCC: 20.5 dBm; 5.2 GHz SRRC/FCC/CE/MIC: 14 dBm; KCC:10 dBm; 5.8 GHz SRRC/FCC: 18 dBm; CE/KCC: 12 dBm
Bluetooth Protocol	Bluetooth 4.2
Bluetooth Operating Frequency	2.4000-2.4835 GHz
Bluetooth Transmission Power (EIRP)	SRRC/MIC/FCC/CE/KCC: 6.5 dBm
GNSS	GPS+GLONASS
Display Device	5.5-inch screen 1920×1090 pixels Brightness: 1000 cd/m² Android System
Random Access Memory, RAM	4GB LPDDR4
Storage Capacity	ROM 32GB + microSD, supports microSD cards with a capacity of up to 128 GB and R/W speed up to UHS-I Speed Grade 3
HDMI	HDMI 1.4
Supported Aircraft	Agras T20
Power Consumption	18 W
Operation Temperature Range	-10° to 40°C
Storage Temperature Range	< 1 month: -30° to 60°C 1-3 months: -30° to 45°C 3-6 months: -30° to 35°C > 6 months: -30° to 25°C (Built-in battery power 40% to 60%)
Charging Temperature	5° to 40°C

REMOTE CONTROLLER - INTERNAL BATTERY

Battery Type	18650 Li-ion (5000 mAh @ 7.2 V)
Working Time	2 hours
Charging Time	2.5 hours (using a standard USB power adapter rated 12 V/2 A)
Power Supply Current/Voltage (USB-A port)	5 V/ 1.5 A
Model	WB37-4920mAh-7.6V

REMOTE CONTROLLER - INTELLIGENT BATTERY

Charge Type	Supports USB power fast charge adapters rated 12 V/2 A
Battery Type	2S LiPo
Capacity	4920 mAh
Voltage	7.6 V
Energy	37.39 Wh
Charging Temperature	5° to 40°C
Model	2 hours

REMOTE CONTROLLER - INTELLIGENT BATTERY CHARGING HUB

Model	WCH2
Input Voltage	17.3-26.2 V
Output Voltage and Current	8.7 V, 6 A
Operating Temperature	5° to 40°C

REMOTE CONTROLLER - INTELLIGENT BATTERY CHARGING HUB POWER ADAPTER

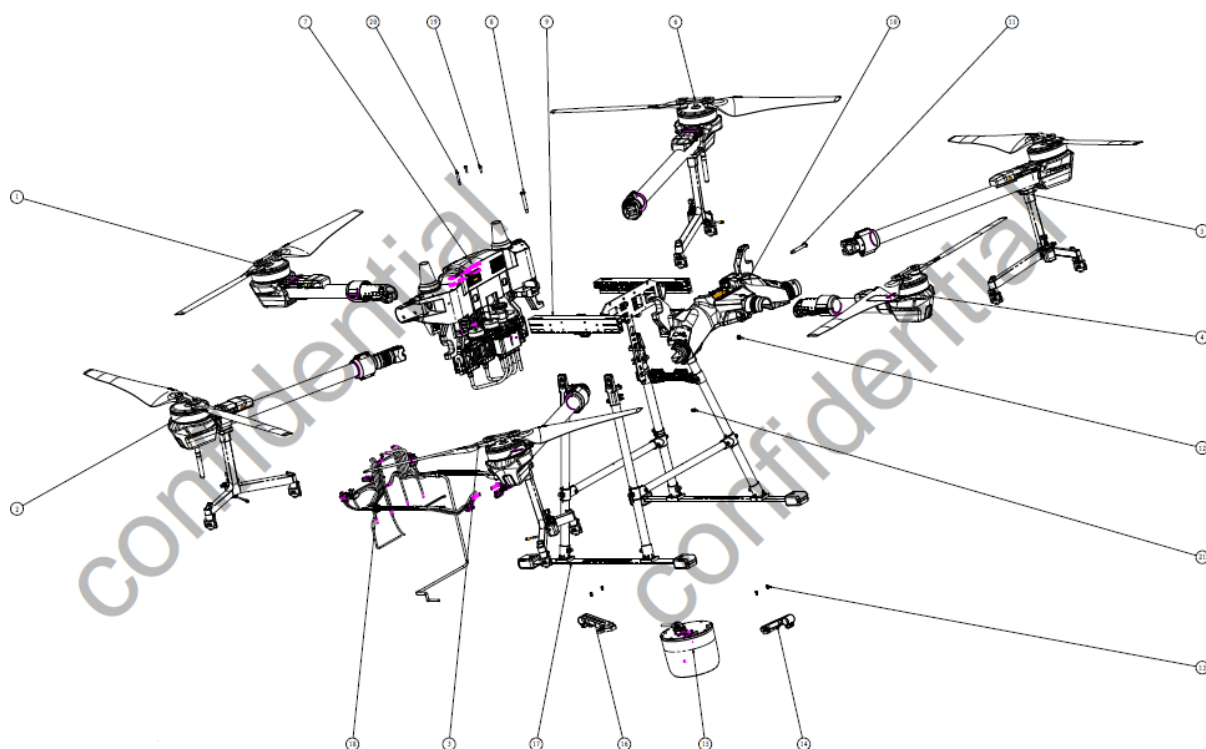
Model	A14-057N1A
Input Voltage	100 - 240 V, 50/60 Hz
Output Voltage	17.4 V
Rated Power	57 W

Remarks

- [1] Support for Galileo will be available at a later date.
- [2] Hovering time acquired at sea level with wind speeds lower than 3 m/s.
- [3] The effective radar range varies depending on the material, position, shape, and other properties of the obstacle.
- [4] Local regulations in some countries prohibit the use of the 5.8 GHz or 5.2 GHz frequencies. In some countries, the 5.2 GHz frequency band is only allowed for indoor use.

2 BOM

2.1 Aircraft

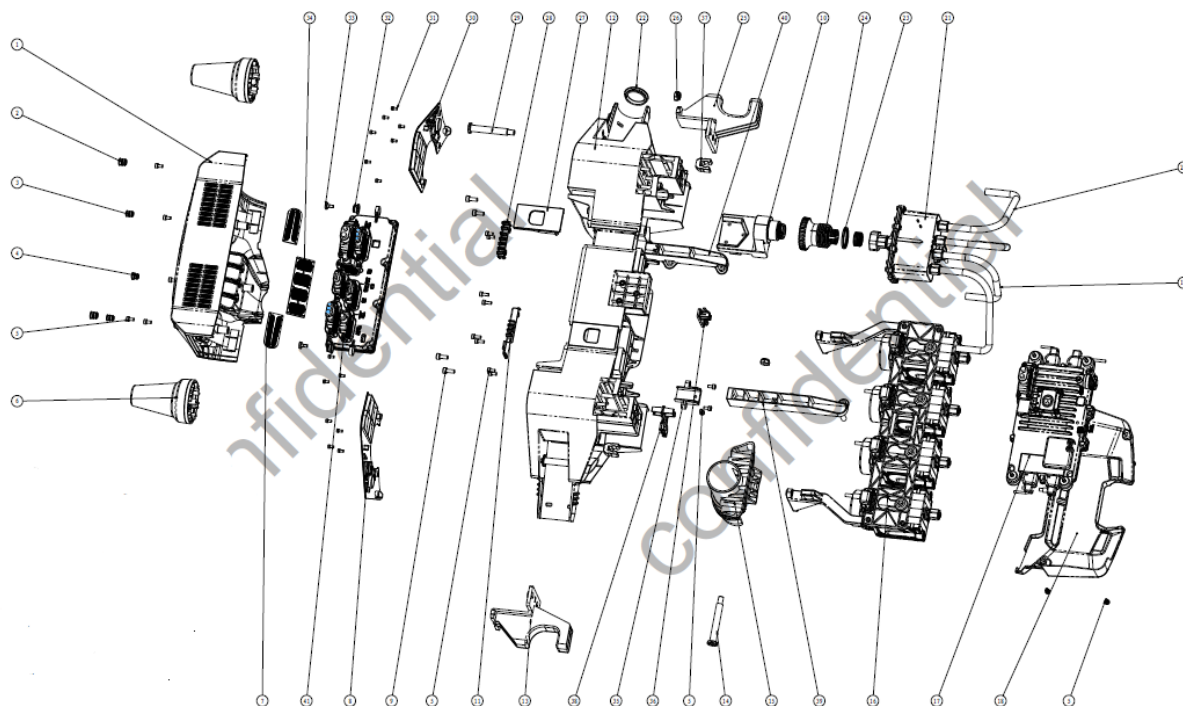


Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	/	M1 Aircraft Arm	1	
2	/	M2 Aircraft Arm	1	
3	/	M3 Aircraft Arm	1	
4	/	M4 Aircraft Arm	1	
5	/	M5 Aircraft Arm	1	
6	/	M6 Aircraft Arm	1	
7	/	Front Frame	1	
8	YC.WJ.ZZ002630.03	Aircraft Arm Pin (Short)	4	
9	/	Middle Frame	1	
10	/	Rear Frame	1	
11	YC.WJ.ZZ002901.03	Aircraft Arm Pin (Long)	2	
12	YC.WJ.LM000044.01	M60-HX100060-5103_01	6	
13	YC.WJ.L00870.02	M30-HC060060-55-85 V2	6	
14	YC.JG.ZS000152.02	Radar Fixing Bracket (Rear)	1	
15	BC.AG.SS000215.01	Omnidirectional Digital Radar	1	
16	YC.JG.ZS000151.02	Radar Fixing Bracket (Front)	1	
17	BC.AG.BB000286.02	Landing Gear	1	
18	/	Cable	1	
19	YC.ST.LL000013.01	M40-HC01400140-070039-0003-Y	16	

20	YC.ST.LL000016.01	M40-HH03250070-080015-0003-Y	4	
21	YC.WJ.L00961.01	M4 Nut	4	

2.2 Front Frame

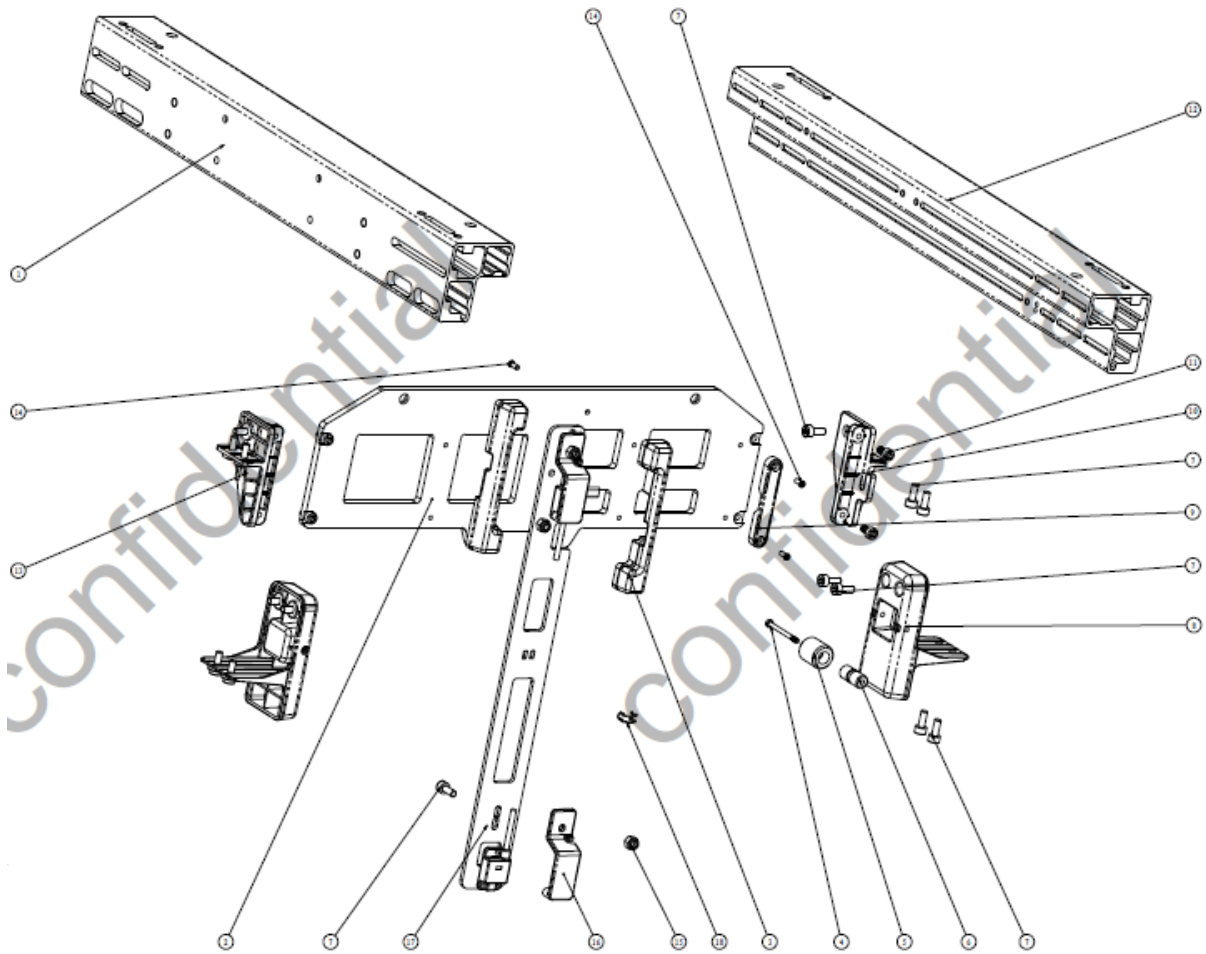


Material List:

No.	Material Number	Material Name	Quantity	Remarks
1&7	BC.AG.SS000240.01	Front Frame Upper Cover Module	1	
2	YC.JG.MY000064.02	Aircraft Head Rubber Stopper (Rear)	2	
3	YC.JG.MY000062.02	Aircraft Head Rubber Stopper (Middle)	2	
4	YC.JG.MY000063.02	Aircraft Head Rubber Stopper (Front)	1	
5	YC.WJ.L00870.02	M30-HC060060-55-85 V2	31	
6	BC.AG.SS000212.01	RTK Antenna Module	2	
7	YC.JG.BJ000065.02	Upper Cover Side Dustproof Mesh	2	
8	YC.JG.ZS000072.03	Front Frame Aircraft Arm Cover (Left)	1	
9	YC.WJ.L01249.02	M40-HC100100-70-12 V2	4	
10	YC.JG.ZS000027.03	Spray Tank Base Plug Port Bottom Base	1	
11	YC.JG.ZS000073.02	Front Frame Cable Holder (Left)	1	
12	BC.AG.SS000248.01	Aircraft Front Frame	1	
13	YC.JG.ZS000078.03	Front Frame Arm Fixing Piece (Left)	1	
14	YC.WJ.ZZ002901.03	Aircraft Arm Pin (Long)	1	
15	YC.SJ.WS002451.02	Front Frame Middle Arm Connector	1	
16	BC.AG.AA000295.02	Delivery Pump and Mounting Bracket	1	
17	BC.AG.SS000209.01	Aerial-Electronics Module (CN)	1	

18	YC.SJ.WS002647.02	Aerial-Electronics Module Protector	1	
19	YC.JG.CS000001.03	4-Channel Flow Meter U-shaped Tube (Short)	2	
20	YC.JG.CS000000.03	4-Channel Flow Meter U-shaped Tube (Long)	2	
21	BC.AG.SS000204.01	4-Channel Electromagnetic Flow Meter	1	
22	YC.XJ.QT000332.01	Aircraft Arm Anti-Scratch Rubber A	3	
23	YC.JG.MY000011.01	Spray Tank Base Filter Sealing Ring	1	
24	YC.JG.ZS000028.02	Spray Tank BASE Plug Port Bottom Base	1	
25	YC.JG.ZS000075.03	Front Frame Arm Fixing Piece (Right)	1	
26	YC.WJ.L00961.01	M4 Nut	3	
27	YC.SJ.WS002453.02	Front Frame Cable Holder Protective Cover	2	
28	YC.JG.ZS000066.01	Front Frame Cable Holder (Right)	1	
29	YC.WJ.ZZ002630.03	Aircraft Arm Pin (Short)	2	
30	YC.JG.ZS000071.03	Front Frame Aircraft Arm Cover (Right)	1	
31	YC.WJ.LL000015	T25-HC050050-40-00	14	
32	YC.JG.MY000020.03	Insulated Rubber Base	2	
33	YC.ST.LL000006.02	M30-HC00980050-080020-5102-Y	2	
34	YC.JG.BJ000066.01	Front Frame Upper Cover Dustproof Mesh	1	
35	YC.WJ.ZZ002789.01	Front Frame Carbon Tube Fixing Piece	2	
36	YC.JG.ZS000025.02	Cable Fixing Piece (Middle)	1	
37	YC.SJ.D00409.02	Hose Clamp 2	2	
38	YC.JG.ZS000024.03	Cable Holder (Front)	2	
39	YC.WJ.ZZ003542.02	Aerial-Electronics Bracket (Left) (Die-Case)	1	
40	YC.WJ.ZZ003541.02	Aerial-Electronics Bracket (Right) (Die-Case)	1	
41	BC.AG.SS000208.01	Spraying Board Module	1	

2.3 Middle Frame

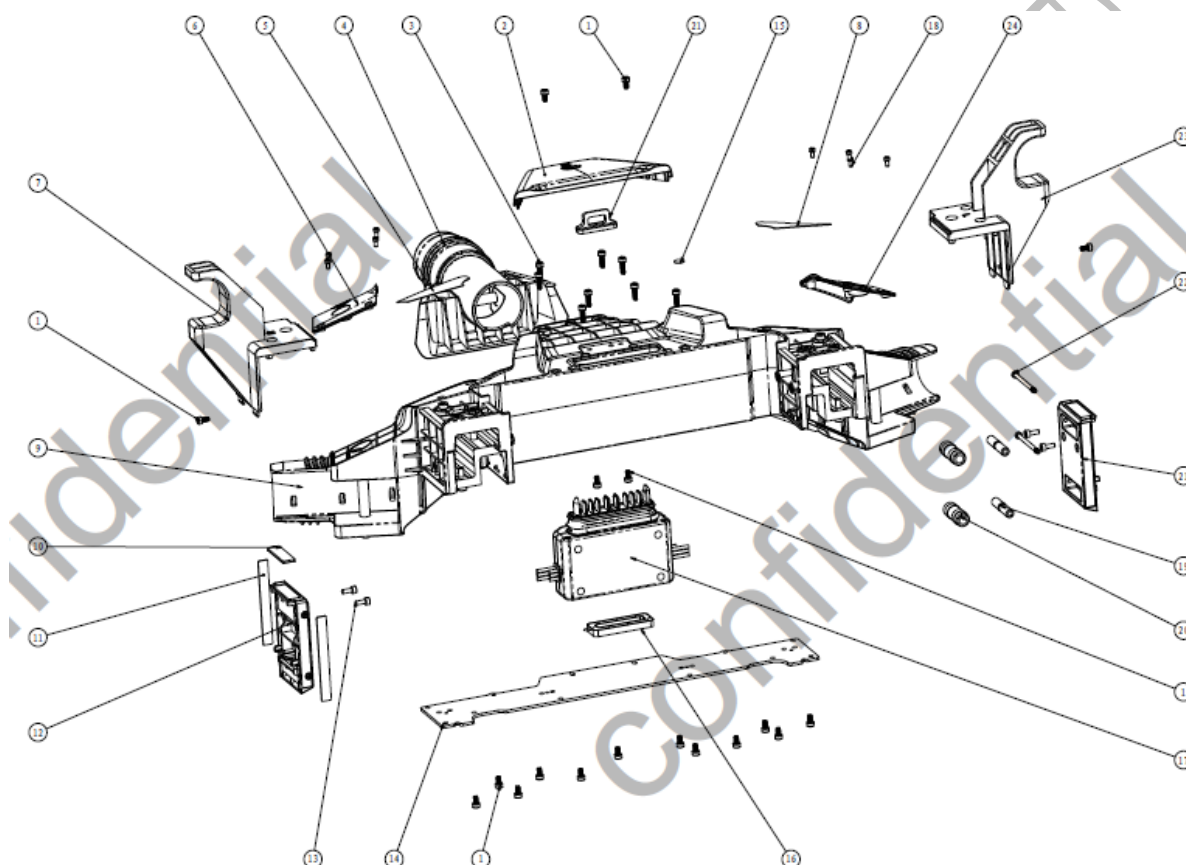


Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	BC.AG.SS000253.01	Airframe L-shaped Aluminum Tube (Right)	1	
2	YC.SJ.TB000089.02	Airframe Fixing Carbon Board	1	
3	YC.SJ.WS002408.03	Spray Tank Rear Guiding Block	2	
4	YC.WJ.ZZ002618.02	Spray Tank Fixing Piece Roller Shaft	2	
5	YC.XJ.QT000247.02	Battery Fixing Piece External Roller Shaft	2	
6	YC.SJ.WS002500.02	Spray Tank Fixing Piece External Roller Shaft	2	
7	YC.WJ.L00868.02	M30-HC080080-55-85 V2	17	
8	YC.SJ.WS002486.02	Spray Tank Fixing Piece	2	
9	YC.SJ.WS002620.02	Spray Tank Limit Cable Clamp	1	
10	YC.SJ.WS003140.01	Middle Frame Carbon Board Fixing Piece (Right)	1	
11	YC.WJ.L00870.02	M30-HC060060-55-85 V2	4	
12	BC.AG.SS000254.01	Airframe L-shaped Aluminum Tube (Right)	1	R Mark
13	YC.SJ.WS003139.01	Middle Frame Carbon Board Fixing Piece (Left)	1	

14	YC.WJ.L01384.00	M20-HC050050-38-85	6	
15	YC.WJ.L00877.00	M3 Nut	3	
16	YC.SJ.WS002410.05	Liquid Meter Sensor Bracket	2	
17	YC.SJ.TB000072.03	Liquid Meter Sensor Carbon Board	1	
18	YC.WJ.TT000182.02	Liquid Meter Cable Holder	1	

2.4 Rear Frame

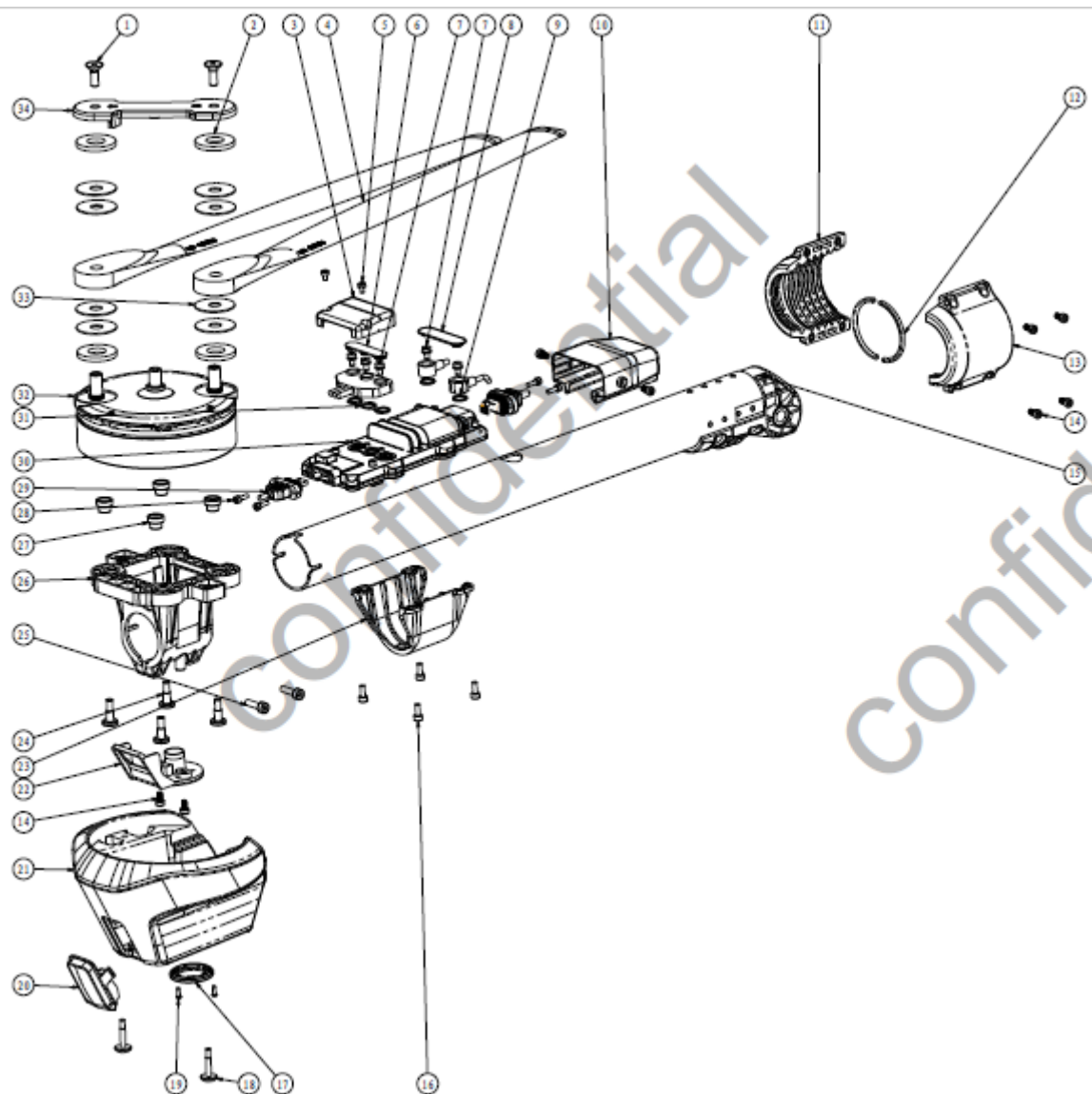


Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	YC.WJ.L00870.02	M30-HC060060-55-85 V2	18	
2	YC.SJ.WS002482.03	Rear Frame_ Middle Arm_Upper Cover	1	
3	YC.WJ.L00871.02	M30-HC100100-55-85 V2	8	
4	YC.SJ.WS002475.04	Rear Frame_ Middle Arm_Connector	1	
5	YC.JD.JD000163.02	Rear Frame Sticker (Right)	1	
6	YC.JG.ZS000069.01	Rear Frame_Side Arm_Upper Cover (Right)	1	
7	YC.JG.ZS000076.03	Rear Frame_Aircraft Arm_Fixing Piece (Right)	1	

8	YC.JD.JD000165.02	Rear Frame Sticker (Left)	1	
9	BC.AG.SS000249.01	Rear Frame	1	
10	YC.XJ.QT000245.04	Battery Fixing Piece_Supporting Pad	2	
11	YC.SJ.MQ000094.01	Rear Frame Teflon Pad	4	
12	YC.SJ.WS002972.03	Rear Frame_ Battery Fixing Piece	2	
13	YC.WJ.L00868.02	M30-HC080080-55-85 V2	4	
14	YC.JG.XW000012.02	Rear Frame_Fixing Fiberglass Board (CN)	1	
15	YC.JD.JD000164.02	Rear Frame Sticker (Round	2	
16	YC.JG.MY000078.01	Power Distribution Board Rubber Damping Pad	1	
17	BC.AG.SS000203.01	Power Distribution Board PCBA (Including ESC Power Cable) (CN)	1	
18	YC.WJ.LL000015	Screw T25-HC050050-40-00	8	
19	YC.SJ.WS002499.02	Battery Fixing Piece Inner Roller	4	
20	YC.XJ.QT000246.01	Battery Fixing Piece External Roller	4	
21	YC.WJ.ZZ003696.01	Aircraft Battery Buckle	1	
22	YC.WJ.ZZ002624.03	Battery Fixing Piece Roller Shaft	4	
23	YC.JG.ZS000077.03	Rear Frame_Aircraft Arm_Fixing Piece (Left)	1	
24	YC.JG.ZS000067.01	Rear Frame_Side Arm_Upper Cover (Left)	1	

2.5 Aircraft Arm

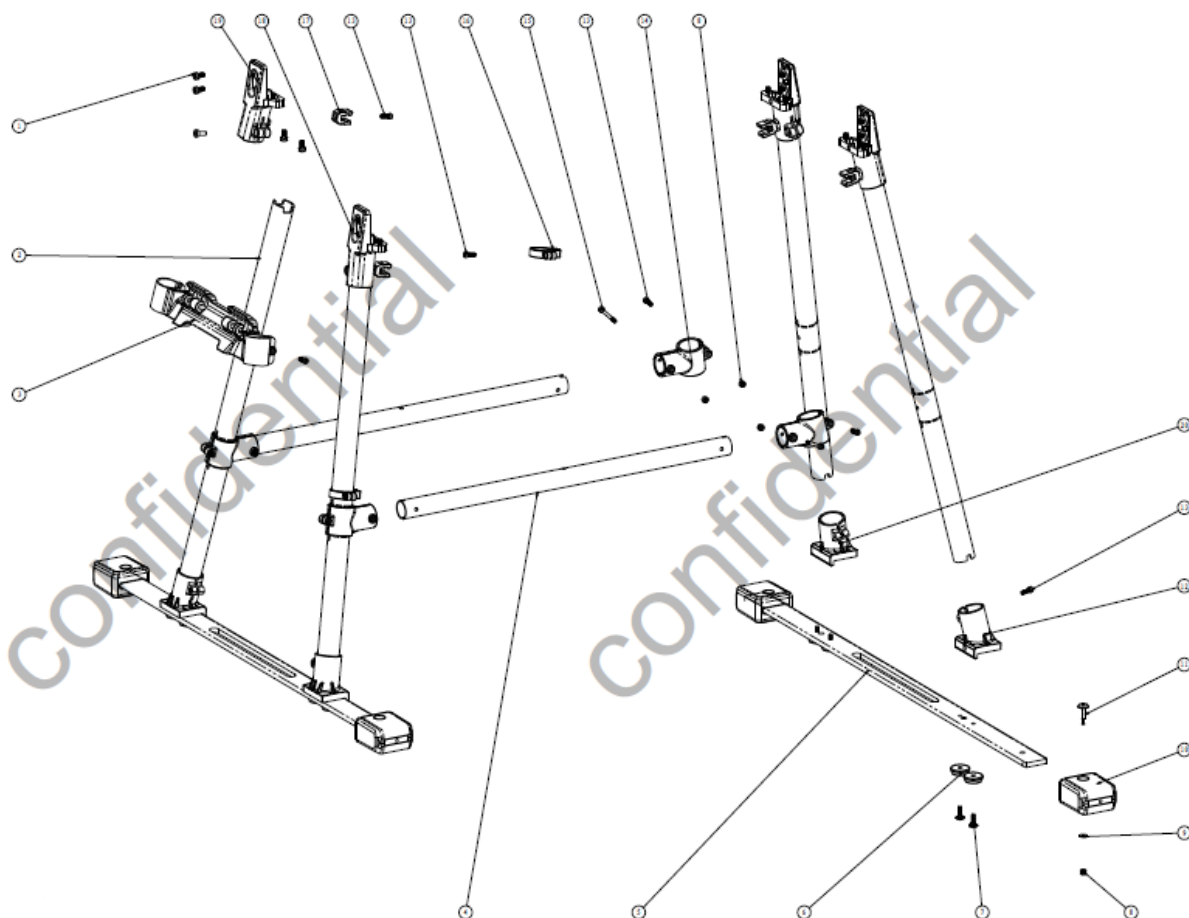


Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	YC.WJ.LL000251.02	M50-HF01200100-116036-5103-Y	2	
2	YC.XJ.QT000196.04	R3185 Propeller Clamp Rubber Gasket	4	
3	YC.SJ.WS002415.03	ESC Upper Shell	1	
4	BC.AG.SS000258.01	3390 Propellers (1 Pair) (CCW)	1	M1/M3/M6
	BC.AG.SS000259.01	3390 Propellers (1 Pair) (CW)	1	M2/M4/M5
5	YC.WJ.LL000138.01	M25-HC00400040-050020-0003-N	2	
6	YC.XJ.QT000336.02	Aircraft Arm Three-Phase Cable Connector Sealing Rubber	1	

7	YC.WJ.LL000244.01	M30-HCW00600060-055020-5103-N_V2	5	
8	YC.XJ.QT000202.03	ESC Cable Locking Screw Sealing Rubber Piece	1	
9	YC.XJ.QT000306.02	ESC Copper Connector Sealing Rubber Piece	2	
10	YC.SJ.WS002831.02	ESC Cover	1	
11	YC.JG.ZS000116.05	Aircraft Arm Locking Sleeve B	1	
12	YC.JG.MY000113.03	Aircraft Arm Locking Sleeve Damping Rubber	2	
13	YC.JG.ZS000117.05	Aircraft Arm Locking Sleeve A	1	
14	YC.WJ.L00870.02	M30-HC060060-55-85 V2	8	
15	BC.AG.SS000251.01	M1&M4 Aircraft Arm (Middle Front/Middle Rear)	1	
15	BC.AG.SS000252.01	M2&M5 Aircraft Arm (Front Left/Rear Right)	1	Long Aircraft Arm
15	BC.AG.SS000250.01	M3&M6 Aircraft Arm (Front Right/Rear Left)	1	Long Aircraft Arm
16	YC.WJ.L00868.02	M30-HC080080-55-85 V2	4	
17	YC.SJ.WS002856.01	LED Board Bracket Plate	1	M1/M3/M4/M5
17	YC.SJ.WS002532.02	SDR Antenna Fixing Base	1	M2/M6
18	YC.WJ.LL000213.02	M30-HT01930060-110025-5103-Y_01	2	
19	YC.WJ.L00536.02	T20PP0600603021	2	
20	YC.SJ.WS002641.02	LED Outer Shell	1	
21	YC.QT.QT000024.05	Motor Shell	1	
22	YC.SJ.WS002640.06	LED Fixing Piece	1	
23	YC.SJ.WS002824.01	Spray Lance Fixing Piece (Short Aircraft Arm)	1	M1/M4
23	YC.SJ.WS002823.01	Spray Lance Fixing Piece (Long Aircraft Arm)		M2/M3/M5/M6
24	YC.WJ.LL000243.02	M40-HC01500075-100020-5103-N_V2	4	
25	YC.ST.LL000013.01	M40-HC01400140-070039-0003-Y	2	
26	YC.JG.ZS000176.01	Motor Fixing Base	1	
27	YC.WJ.ZZ003448.02	Motor Locating Piece	4	
28	YC.WJ.L01148.00	M2.5*8 Screw	4	
29	YC.SJ.WS002524.01	Waterproof Plug	2	Belongs to ESC Composite Cable
30	BC.AG.SS000106.01	ESC Module	1	
31	YC.XJ.QT000338.03	Motor Three-Phase Cable Waterproof Three-Jointed Rubber Ring	1	
32	YC.SJ.WS002379.01	Motor Rotor Shell	2	
33	YC.SJ.WS002416.03	R3185 Propeller Clamp Telfon Pad	8	

2.6 Landing Gear



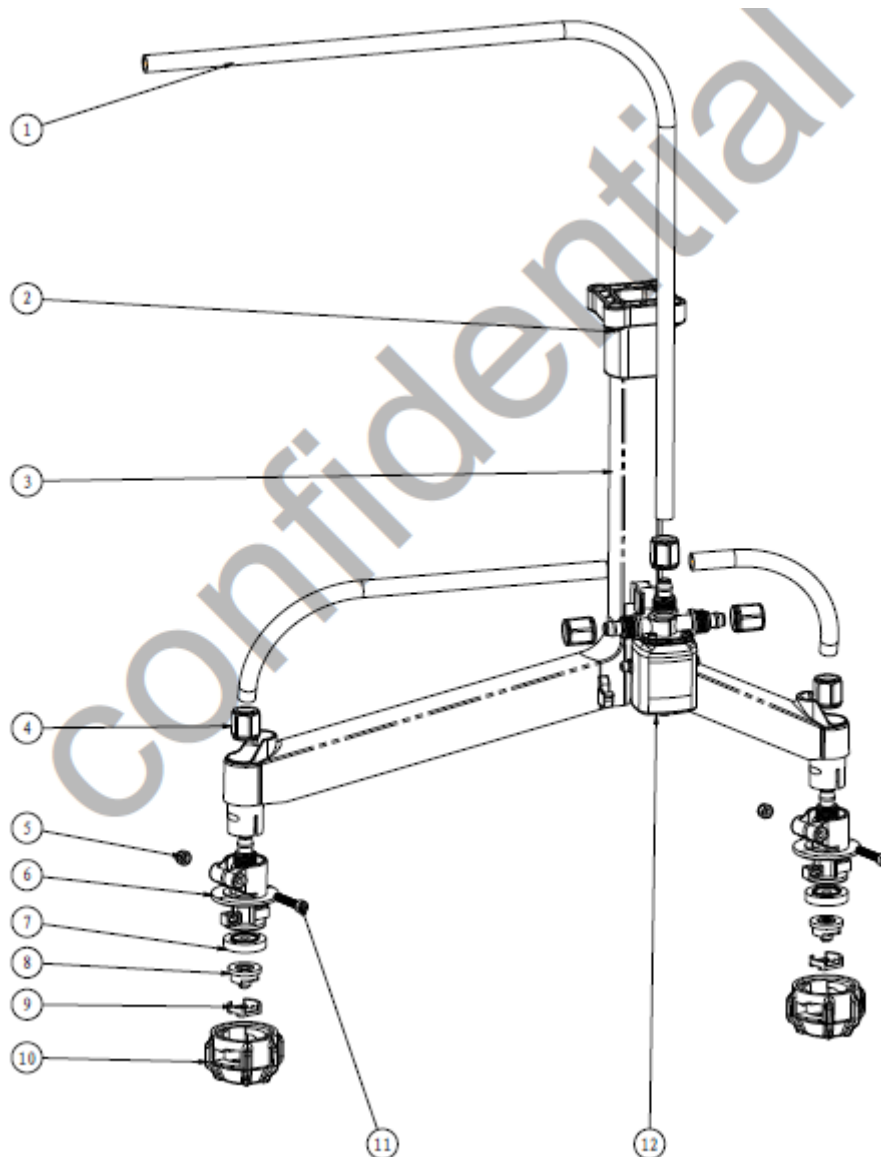
Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	YC.WJ.LL000232.01	M40-HC01000100-072028-5103-N_01	20	
2	BC.AG.SS000256.01	Landing Gear Vertical Rod	4	
3	YC.JG.ZS000271.02	Landing Gear_Remote Controller Mounting Piece	1	
4	BC.AG.SS000255.01	Landing Gear Bottom Fiberglass Board	2	
5	YC.JG.XW000017.01	Landing Gear Crossbeam (Radar Fixing Pole)	2	
6	YC.XJ.QT000370.02	Landing Gear Buffer Pad	8	
7	YC.WJ.LL000250.01	M30-HPW01200100-090025-5103-Y_01	8	
8	YC.WJ.L00877.00	M3 Nut	12	
9	YC.WJ.Z01729.01	Gasket	4	
10	YC.JG.FP000001.02	Landing Gear Fiberglass Rubber Sleeve	4	

Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	YC.SJ.WS003584.04	20L Spray Tank	1	
2-10	BC.AG.SS000124.01	Spray Tank Liquid Lever Meter	2	
11	YC.SJ.WS002434.02	Spray Tank Bottom Part (Front Right/Rear Left)	2	
12	YC.XJ.QT000198.01	Spray Tank Bottom Cover_Sealing Pad	1	
13	YC.SJ.WS002420.04	Spray Tank Bottom Cover	1	
14	YC.SJ.WS002437.02	Spray Tank Bottom Part (Front Left /Rear Right)	2	
15	YC.WJ.L00870.02	M30-HC060060-55-85 V2	8	
16	YC.XJ.QT000199.02	Spray Tank Outlet Sealing Pad	1	
17	YC.SJ.WS002417.02	Spray Tank Outlet	1	
18	YC.SJ.WS002422.02	Spray Tank Outlet Nut	1	
19	YC.SJ.WS002419.02	Spray Tank Hose	0.28	M
20	YC.SJ.J01186.02	Filter Rubber Ring	1	
21	YC.XJ.QT000422.01	Spray Tank 100 Holes Filter Sealing Ring	1	
22	YC.SJ.W01124.01	Spray Tank 100 Holes Filter	1	
23	YC.SJ.W01123.01	Spray Tank 50 Holes Filter	1	
24	YC.XJ.QT000204.04	Spray Tank One-Way Valve External Shell Sealing Ring	1	
25	YC.SJ.WS002436.04	Spray Tank One-Way Valve External Shell	1	
26	YC.JG.MY000026.02	One-Way Valve Core Sealing Ring	1	
27	YC.SJ.WS003561.02	Spray Tank One-Way Valve Core	1	
28	YC.WJ.TT000157.04	One-Way Valve Core Spring	1	
29	YC.SJ.WS003560.01	One-Way Valve Core Limit Block	1	
30	YC.JG.MY000025.01	One-Way Valve Bottom Base Sealing Ring	1	
31	YC.SJ.WS002441.05	One-Way Valve Bottom Base	1	
32	YC.XJ.QT000207.01	Spray Tank One-Way Valve Stopper Sealing Ring	1	
33	YC.WJ.L00868.02	M30-HC080080-55-85 V2	2	
34	YC.SJ.WS002443.02	Spray Tank One-Way Valve Side Stopper	1	
35	YC.WJ.LL000211.02	M30-HC02050080-070020-5103-Y_V2	1	
36	YC.SJ.WS002504.04	Spray Tank Inlet Filter	1	
37	YC.SJ.WS002423.02	Spray Tank Fixing Ring	1	
38	YC.SJ.WS002421.02	Spray Tank Cover	1	
39	YC.XJ.QT000200.02	Spray Tank Cover Ventilation Sealing Pad	1	
40	YC.XJ.QT000197.02	Spray Tank Cover Sealing Pad	1	
41	YC.SJ.WS002418.04	Spray Tank Top Cover	1	
42	YC.WJ.ZZ003452.01	Figure 8 Cable Clamp	2	
43	YC.QT.QT000055.01	Spray Tank Top Cover Lanyard	0.14	M

2.8 Spraying System Module (to be determined)



Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	YC.SJ.WS001431.03	White Spraying Nylon Tube	1	
2	YC.WJ.LL000120	M30-HPW080050-80-85	4	
3	YC.JG.ZS000182.03	Sprinkler Extend Fixing Rod (Left)	1	M2/M3
3	YC.JG.ZS000181.03	Sprinkler Extend Fixing Rod (Right)	1	M5/M6
4	YC.SJ.L00006.01	Nut	5	
5	YC.WJ.L00877.00	M3 Nut	2	
7	YC.SJ.J00794.03	Sprinkler Sealing Ring	2	
8	YC.SJ.WS002723.02	Nozzle	2	
9	YC.SJ.J00790.02	Sprinkler Locating Ring	2	

Material List:

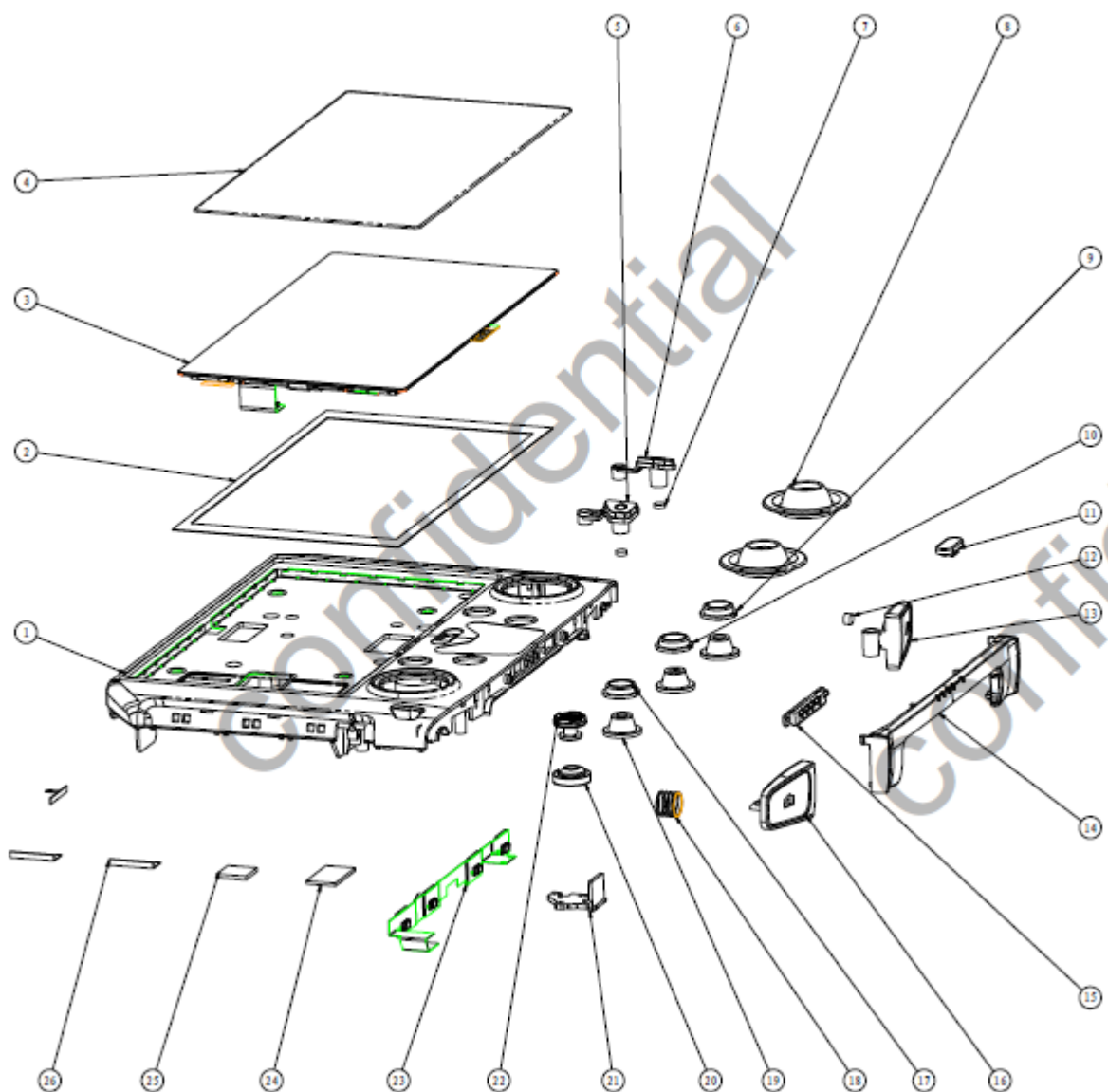
No.	Material Number	Material Name	Quantity	Remarks
1	YC.JG.MQ000171.03	Upper Cover_Bottom Decorative Piece	1	
2	YC.JG.MY000075.02	Remote Controller Soft Rubber (Left)	1	
3	YC.JG.MY000074.02	Remote Controller Soft Rubber (Right)	1	
4	XN.AG.000076	Lower Shell Module (Virtual Layer)	1	
5	YC.PM.QT000065.01	Battery EVA Foam	2	
6	YC.DZ.BB000052.01	Remote Controller Built-in Battery	1	
7	YC.BZ.TZ000106.01	Fan Cable Fixing Sticker	1	
8	YC.JG.MQ000247.01	Fan Bracket_Compressing Foam	1	
9	YC.SJ.WS001879.02	Image Transmission Antenna Block	1	
10	BC.MA.SS000069.01	SDR Board	1	
11	BC.RM.PP000087.04	Left Dial Board (RH)	1	
12	BC.RM.PP000089.04	Right Dial Board (RH)	1	
13	YC.SJ.MQ000098.02	Round Rubber Pad	2	
14	YC.SJ.WS001872.02	Main Board Fixing Board	2	
15	YC.WJ.L00655.02	M20-PP050050-30-21 V2	6	
16	BC.AG.SS000233.01	Remote Controller Main Board	1	
17	YC.WJ.L00018.04	IMU Screw M1.6*3.5	4	
18	YC.WJ.L01124.01	M1.6*3 Screw	10	
19	YC.PM.QT000064.01	Fixing Piece Foam	5	
20	YC.SJ.MQ000051.01	Battery Cable Fixing Piece	2	
21	YC.WJ.RR000059.04	Heat Sink	1	
22	YC.WJ.L00158	Screw (3PUTB)φ2.0×5.0	43	
23	YC.PM.DD000038.01	Heat Sink Thermal Foam	2	
24	BC.MA.SS000071.01	Right Control Stick	1	
25	BC.MA.SS000070.01	Left Control Stick	1	
26	YC.WJ.L01523.01	T20-PP065065-35-11	8	
27	XN.AG.000074	Upper Shell (Virtual Layer)	1	
28	YC.WJ.ZZ002098.01	Control Sticker Head	2	
29	YC.SJ.JG000102.01	Control Sticker Soft Rubber Ring	2	
30	YC.DZ.PF000240.03	Left Hall Sensor Board Flat Cable	1	
31	BC.RM.PP000082.04	Record Button Board	1	
32	YC.DZ.PF000234.03	Flat Cable Connecting the Main Board and Button Board	1	
33	YC.DZ.PF000239.03	Right Hall Sensor Board Flat Cable	1	
34	BC.RM.PP000083.04	Focus/Shutter Button Board	1	
35	BC.RM.PP000086.03	SDR Board Flat Cable	1	
36	XN.AG.000075	Front Cover & Antenna Module (Virtual Layer)	1	
37	YC.DZ.AA000081.03	Wi-Fi Antenna 2	1	

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38	YC.DZ.AA000080.02	Wi-Fi Antenna 1	1	
39	YC.BZ.J00931.01	Flat Cable Connector Sticker	2	
40	BC.RM.PP000084.05	Photosensitized Board	1	
41	YC.PM.QT000043.03	Heat Sink Shielding Piece	1	
42	YC.DZ.AS000004	Speaker	1	
43	YC.JG.MQ000245.01	Remote Controller Foam,	1	
44	YC.PM.DD000036.01	Main Board Thermal Foam	1	
45	YC.JG.ZS000173.03	Fan Bracket	1	
46	YC.DZ.BB000052.01	Remote Controller Built-in Battery	1	
47	YC.JG.MQ000246.01	Remote Controller Fan Foam	1	
48	YC.JG.MY000152.01	Shielding Piece	1	
49	YC.JG.FM000003.02	Remote Controller Lanyard	2	

2.10 Remote Controller Upper Cover



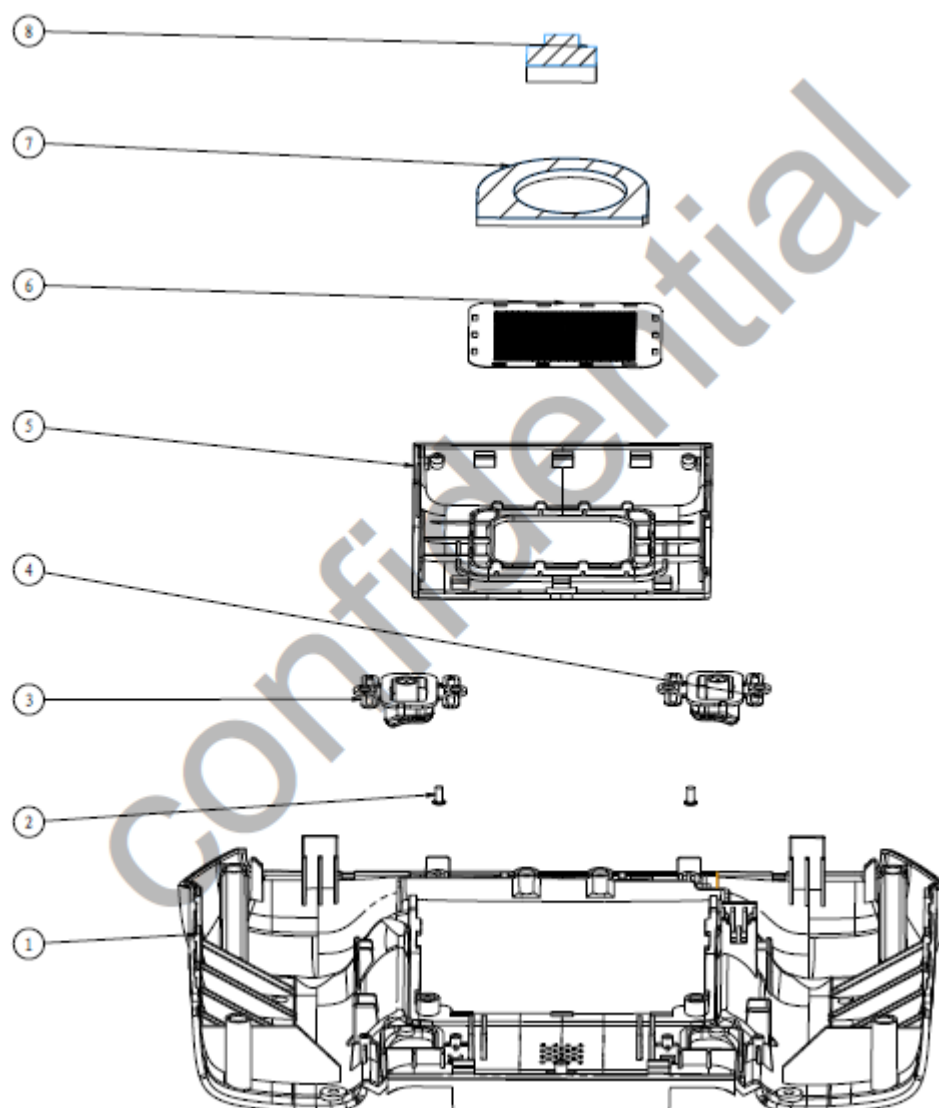
Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	BC.AG.SS000238.01	Upper Shell Module	1	
2	YC.JD.JD000064.03	Screen Back Adhesive	1	
3	YC.DZ.EE000018	Touchscreen	1	
4	YC.OC.PG000001.02	Screen Tempered Glass Protective Film	1	
5	YC.SJ.WS001877.03	Confirm Button / Custom Button C3	1	
6	YC.SJ.WS001876.03	Return Button/Function Button	1	
7	YC.XJ.QT000110.01	Round Gasket	2	
8	YC.JG.MY000076.02	Control Stick Dustproof Ring	2	
9	YC.SJ.WS001871.03	RTH Button	1	
10	YC.JG.ZS000369.01	Pause Button	1	
11	YC.SJ.WS001883.03	Flight Mode Switch	1	
12	YC.XJ.QT000177.01	Record Button Gasket	1	
13	YC.SJ.WS001882.04	Record Button	1	
14	YC.SJ.WS001859.02	Front Cover Decorative Piece	1	
15	YC.SJ.WS001864.02	Light Guide Beam	1	
16	YC.SJ.WS001875.04	Shutter Button	1	
17	YC.SJ.WS001880.03	Power Button	1	
18	YC.WJ.TT000140.01	Shutter Button Spring	1	
19	YC.XJ.QT000108.02	Upper Shell Button Silicone Rubber Pad	3	
20	YC.XJ.QT000106.03	5D Button Decorative Silicone Rubber	1	
21	YC.SJ.WS001984.03	Button Board Bracket	1	
22	YC.SJ.WS001881.03	5D Button	1	
23	BC.AG.SS000243.01	Microphone Button	1	
24	YC.PM.DD000042.01	GPS Foam	1	
25	YC.PM.DD000037.01	Screen Thermal Foam	1	
26	YC.SJ.WS002747.01	Photosensitized Shielding Piece	1	

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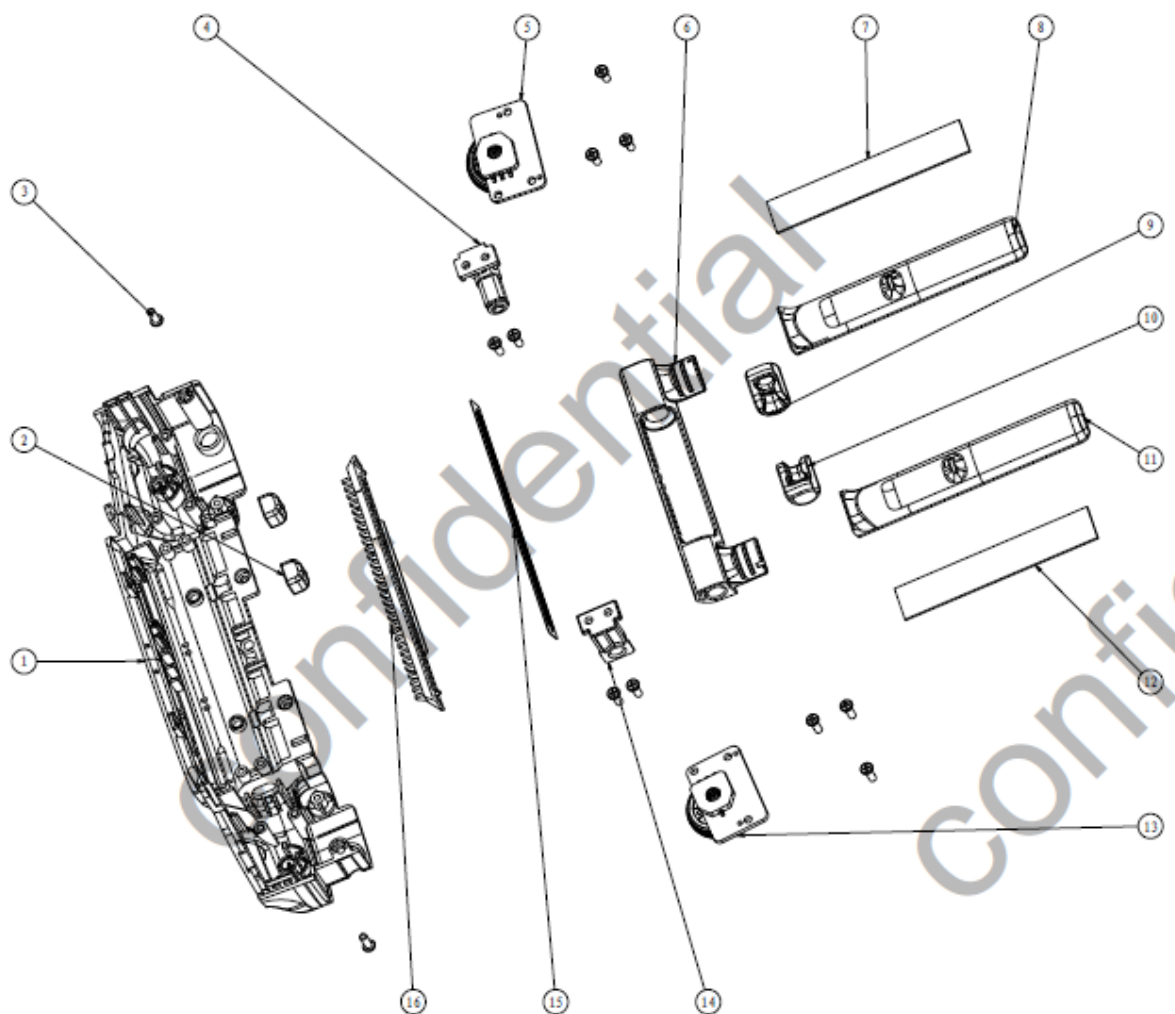
2.11 Remote Controller Lower Cover



Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	BC.AG.SS000236.01	Lower Shell Module	1	
2	YC.WJ.L01491.02	T16-PP036036-28-11 V2	2	
3	YC.SJ.WS001876.03	Return Button/Function Button	1	
4	YC.SJ.WS001863.02	Custom Button C1	1	
5&6	BC.AG.SS000235.01	Remote Controller Rear Cover	1	
7	YC.JG.MQ000113.02	Air Inlet Cover Board Dustproof Foam	1	
8	YC.JG.MQ000253.01	MIC Board Thermal Foam	1	

2.12 Remote Controller Front Cover

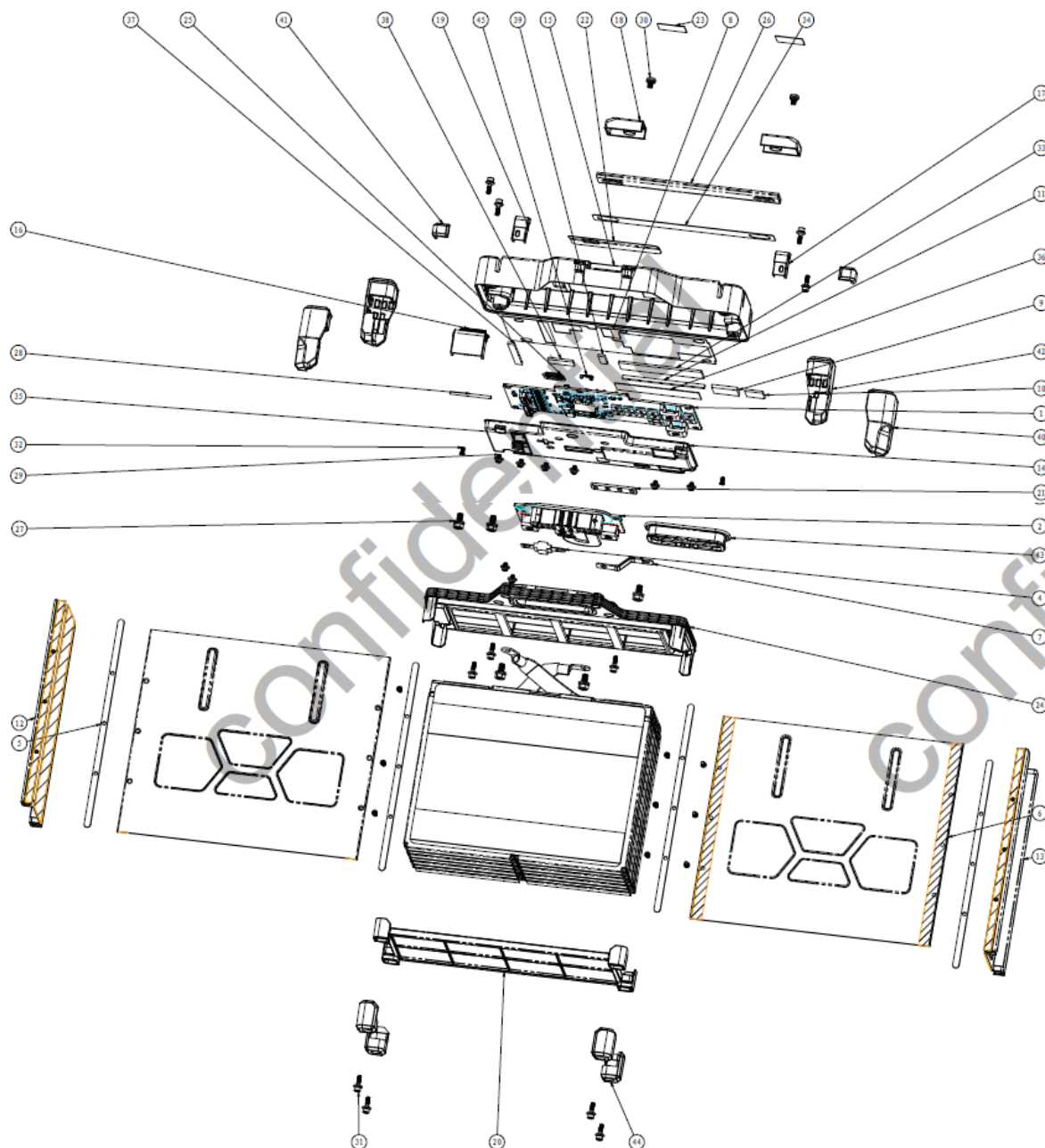


Material List:

No.	Material Number	Material Name	Quantity	Remarks
1	YC.JG.ZS000229.01	Remote Controller Front Cover	1	
2	YC.JG.MY000086.01	Front Cover Rubber Stopper	2	
3	YC.WJ.L01124.01	M1.6*3 Screw	12	
4	YC.WJ.CC000093.02	Antenna Angle Shaft	1	
5	BC.MA.SS000068.01	Right Dial Module	1	
6&9&10	BC.MA.SS000084.01	Antenna Base	1	
7	YC.DZ.AA000083.03	Image Transmission Antenna (Left) (Short)	1	
8	YC.SJ.WS001857.03	Antenna Right Sleeve	1	
9	YC.XJ.QT000156.04	Control Stick Storage Slot Silicone Rubber (Right)	1	
10	YC.XJ.QT000157.04	Control Stick Storage Slot Silicone Rubber (Left)	1	
11	YC.SJ.WS001862.03	Antenna Left Sleeve	1	
12	YC.DZ.AA000084.03	Image Transmission Antenna (Right) (Long)	1	

13	BC.MA.SS000067.01	Left Dial Module	1	
14	YC.SJ.WS001861.02	Antenna Shaft	2	
15	YC.WJ.SZ000014.02	Air Outlet Dustproof Mesh	1	
16	YC.SJ.WS001865.04	Front Cover Air Outlet Decorative Piece	1	

2.13 Battery



Material List

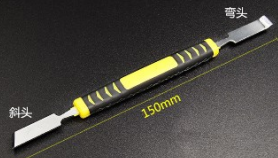
No.	Material Number	Material Name	Quantity	Remarks
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1	BC.AG.SS000296.01	Battery Management Board PCBA	1	
2	BC.AG.SS000297.01	Battery Port Board PCBA	1	
3	YC.DZ.UH000014	Electric Current Fuse	1	
4	YC.JG.BJ000200.01	Negative Electrode Conductive Copper Piece	1	
5	YC.JG.MQ000198.02	Protective Board Anti-Static Sticker	1	
6	YC.JG.MQ000200.01	Protective Board Heat Sink Thermal Conductive Pad	1	
7	YC.JG.MQ000228.01	Protective Board U-shaped Copper Rod Thermal Conductive Pad	1	
8	YC.JG.MQ000229.01	Conductive Copper Rod Thermal Conductive Pad	1	
9	YC.JG.XS000000.01	Protective Board Vacuum Thermoform Cover	1	
10	YC.JG.YZ000044.01	Battery Upper Cover	1	
11	YC.JG.YZ000045.02	Battery Front Clamp	1	
12	YC.JG.YZ000046.01	Battery Front Clamp Fixing Base_Left	1	
13	YC.JG.YZ000047.01	Battery Handle Upper Cover	1	
14	YC.JG.YZ000048.01	Battery Front Clamp Fixing Base_Right	1	
15	YC.PM.QT000124.02	Battery Level Indicator Shielding Foam	1	
16	YC.SJ.MQ000066.03	Battery Level Sticker	1	
17	YC.SJ.MQ000068.01	Battery Handle Upper Cover Paste	1	
18	YC.SJ.WS002376.02	Battery Button	1	
19	YC.SJ.WS002377.01	Battery Handle	1	
20	YC.ST.LL000029.01	M50-PH01350070-086035-0003-Y	5	
21	YC.WJ.CC000139.01	Battery Front Clamp Rotation Axis	1	
22	YC.WJ.LL000226.01	M30-PBW00700030-055020-3123-Y_01	8	
23	YC.WJ.LL000227.01	M40-PBW00900060-080030-5103-Y01	2	
24	YC.WJ.LL000229.02	M30-PF00600060-050000-0220-Y_V2	2	
25	YC.WJ.RR000068.01	Protective Board Heat Sink	1	
26	YC.WJ.TT000165.03	Battery Handle Inner Liner	1	
27	YC.WJ.TT000167.02	Battery Front Clamp Spring	1	
28	YC.XJ.DR000014.01	Protective Board Thermal Pad - 0101	1	
29	YC.XJ.DR000015.01	Protective Board Thermal Pad - 0301	1	
30	YC.XJ.DR000016.01	Protective Board Thermal Pad - 0201	1	
31	YC.XJ.DR000017.01	Protective Board Thermal Pad	1	
32	YC.XJ.JZ000012.01	Battery Upper Cover Front Pad	2	
33	YC.XJ.QT000185.03	Battery Connector Waterproof Ring	1	

34	YC.XJ.QT000277.01	Button Soft Rubber Pad	1	
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3 Disassembly and Assembly & Notices

3.1 Tool

Tool	Specifications/Picture
3.0 mm Hexagonal Socket Screwdriver	
2.5 mm Hexagonal Socket Screwdriver	
2.0 mm Hexagonal Socket Screwdriver	
1.5 mm Hexagonal Socket Screwdriver	
Tweezers (Metal)	
P00 Screwdriver	
Long Nose Pliers	
Anti-Static Tweezers	
Crowbar	

Cutting Pliers	
10/7mm Sleeve	
RF Torque Wrench	
Flow Meter Hexagonal Nut Fixture	
Hose Quick-Release Nut Fixture	
Rubber Hammer	

3.2 Video Links

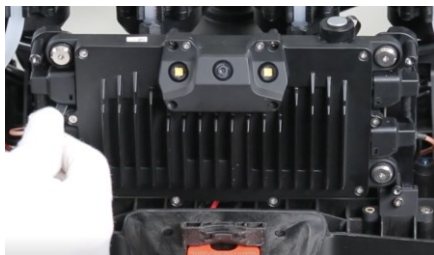
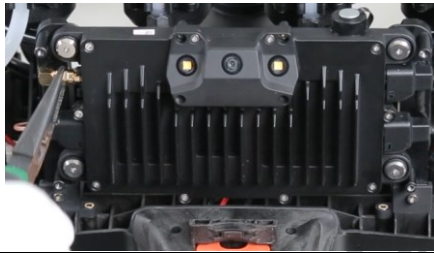
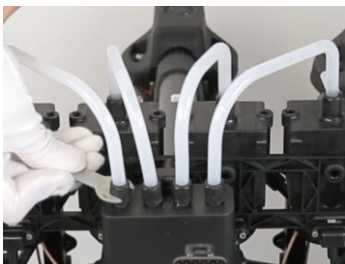
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Disassembling the T20 Aircraft	https://cdn.djivideos.com/watch/d672ae3d-ea5c-422c-aa8c-225c2a677295?autoplay=false&poster=	
Assembling the T20 Aircraft	https://pan-sec.djicorp.com/s/7WXTStctrpanBao	gkas123
Disassembling the T20 Remote Controller	https://cdn.djivideos.com/watch/c6379b8d-a0b9-4bdc-8f82-948c97b8844a?autoplay=false&poster=	
Assembling the T20 Remote Controller	https://cdn.djivideos.com/watch/0f9e9b2f-2ea6-438e-8c81-2bc817832034?autoplay=false&poster=	
Disassembling the 4-Channel Charger	https://cdn.djivideos.com/watch/130c42de-10df-4e2b-8fa4-17d898d93474?autoplay=false&poster=	
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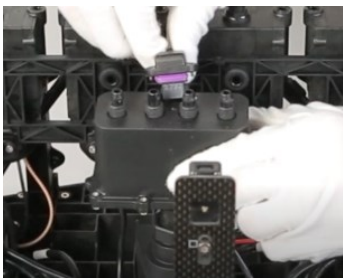
3.3 Disassembly and Assembly Steps

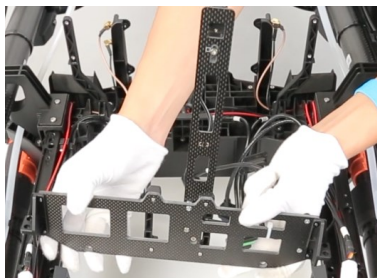
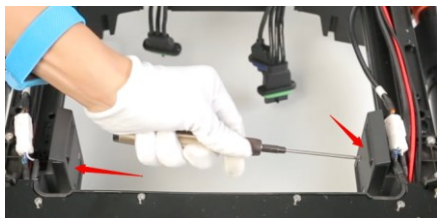
3.3.1 Whole Product

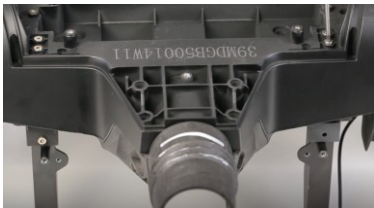
No.	Disassembly and Assembly	Picture	Notice
1	Remove the propeller holder on the M1 aircraft arm.		
2	Remove the two screws securing the upper propeller adapter on the M1 aircraft arm, then remove the propeller adapter, gaskets, and propellers. Remove the propeller adapters, gaskets, and propellers on other aircraft arms in the same way.		When mounting the propellers, distinguish between CCW and CW propellers.
3	Use the tweezers to remove the five rubber stoppers on the screws of the upper cover on the front frame.		
4	Remove the five screws securing the upper cover on the front frame. Then remove the upper cover module.		

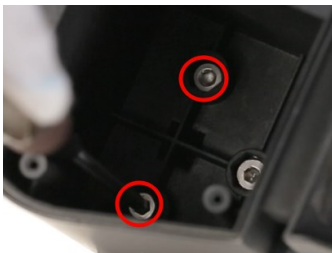
5	Remove the cables from the cable holders as shown.		
6	Remove the screws securing the cables on the spraying board module, then disconnect the cables. Remove the cables from the cable holders as shown.		
7	Remove the two screws securing the spraying board module, then remove the spraying board module.		
8	Remove the seven screws securing the front frame arm cover (left), then remove the front frame arm cover (left). Then do the same way to remove the right aircraft arm cover.		
9	Remove the four screws securing the protective cover of the aerial-electronics module, then remove the cover.		

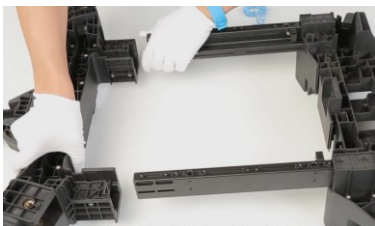
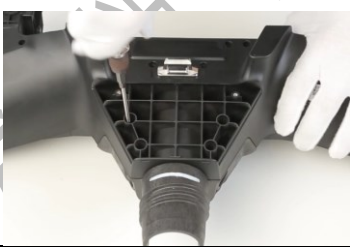
10	Pry off the fixing piece of the antenna connector on the aerial-electronics module.		
11	Screw out the feeder connector.		During assembly, note that the SDR antenna port is different with the RTK antenna port.
12	Remove the four screws securing the aerial-electronics module, then remove the aerial-electronics module.		
13	Remove the nut on the quick-release connector of the water outlet, then remove the hose.		
14	Remove the eight screws securing the right landing gear supporting rods on the landing gear module. Remove the hoses and landing gear module.		
15	Screw out the nuts and remove the 4-channel flowmeter U-type hose.		

16	Remove the two screws securing the liquid meter signal cable, then pull out the cable connector.		
17	Loosen the nut of the quick-release connector on the 4-channel flowmeter, then remove the flowmeter.		
18	Remove the spray tank base plug port (round ridged base).		
19	Remove the three screws securing the bottom base of the spray tank plug port, then remove the bottom base.		
20	Remove the four screws securing the delivery pump mounting bracket, then remove the delivery pump and delivery pump mounting bracket.		
21	Remove the eight screws securing the left and right spray tanks' fixing parts, then remove the fixing parts.		

22	Remove the six screws securing the left and right airframe carbon boards' fixing parts, then remove the liquid level sensor module.		
23	Remove the four screws securing the left and right batteries' fixing parts, then remove the fixing parts.		
24	Remove the 12 screws securing the rear frame fixing fiberglass board, and then remove the fiberglass board.		
25	Remove the M1 ESC signal cables' connectors and power cable.		
26	Loosen the nut securing the M1 aircraft arm screw bolt with the 10 mm sleeve, then remove the screw bolt.		Long screw bolt is used for M1/M4 aircraft arm.
27	Remove the M1 aircraft arm. Remove the other five aircraft arms in the same way.		Pay attention to the installation position of each aircraft arm

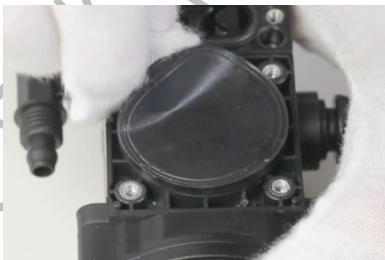
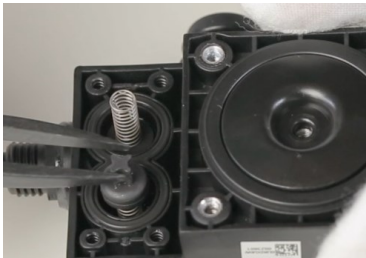
28	Remove the four screws securing the power distribution board module on the rear frame.		
29	Remove the four screws securing the delivery pump brackets (left and right), then remove the delivery pump brackets (left and right).		
30	Turn over the drone as shown. Remove the two screws securing the cable fixing pieces (left, middle and right), then remove the cable fixing pieces (left, middle and right).		
31	Remove the three screws securing the aerial-electronics bracket's fixing piece.		
32	Then remove the two screws securing the aerial-electronics bracket. Remove the aerial-electronics bracket's fixing piece and the aerial-electronics bracket.		

33	Remove the three screws securing the RTK antenna module, then remove the RTK antenna module.		A long screwdriver is required to remove the screw.
34	Remove the eight screws securing the aircraft arm's fixing pieces (left and right) on the front frame, then remove the aircraft arm's fixing piece (left and right).		
35	Turn over the drone as shown. Remove the six screws securing the aircraft arm's fixing pieces (left and right) on the rear frame.		
36	Next, remove the screw bolts with the sleeve and remove the aircraft arm's fixing pieces (left and right) on the rear frame.		7mm Sleeve
37	First, remove the four screws securing the airframe L-type aluminum tubes (left and right); next, remove the screw bolts with the sleeve.		
38	Remove the four screws securing the airframe L-type aluminum tubes (both left and right).		

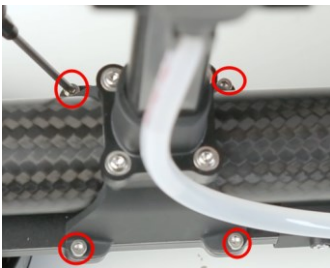
39	Separate the airframe L-type aluminum tubes (both left and right) from the front frame and rear frame.		
40	Remove the four screws securing the middle arm's connector on the front frame, then remove the connector.		
41	Remove the two screws securing the middle arm's upper cover on the rear frame, then remove the upper cover.		
42	Remove the four screws securing the middle arm's connector on the rear frame, then remove the connector.		
43	Remove the eight screws securing the side arm's upper covers (left and right) on the rear frame, then remove the upper covers (left and right).		

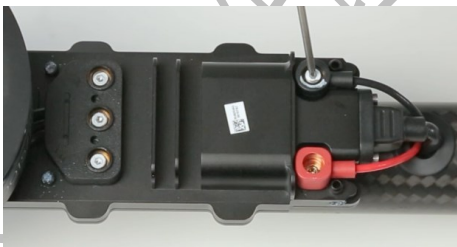
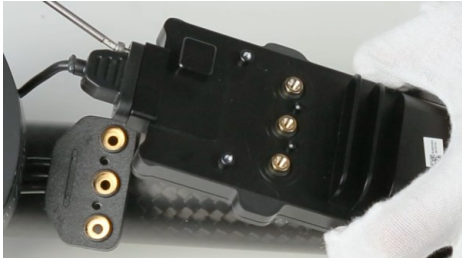
3.3.2 Disassembling the Delivery Pump

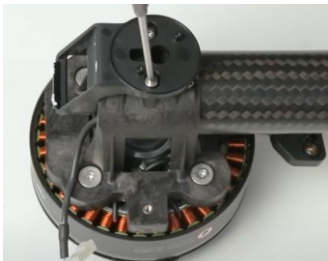
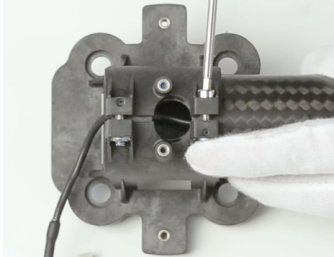
No.	Disassembly and Assembly	Picture	Notice
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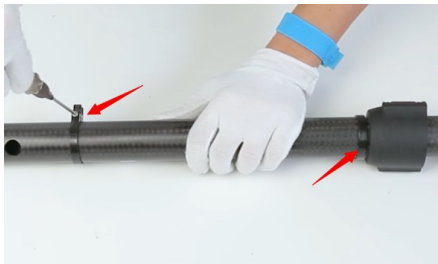
1	Remove the rubber on the delivery pump motor.		
2	Remove the screw securing the hose buckle, then remove the buckle.		
3	Remove the six screws securing the delivery pump's cover, then remove the cover.		
4	Rotate the delivery pump's rubber diaphragm to remove it.		
5	Remove the four screws securing the delivery pump valve base, then separate the sealing ring and valve base.		
6	Remove the two spring rods on the one-way valve and two springs. In the same way, remove the other sealing ring and valve base on the delivery pump valve base. Remove the two spring rods on the one-way valve and two springs.		

3.3.3 Disassembling the Aircraft Arm

No.	Disassembly and Assembly	Picture	Notice
1	Remove the two screws securing the SDR antenna fixing base, then remove the fixing base.		
2	Remove the SDR antenna.		
3	Remove the two screws securing the motor protective cover, then remove the cover.		
4	Remove the insulation sticker and disconnect the electromagnetic relief valve cable.		
5	Remove the four screws securing the spray lancer module and pull out the electromagnetic relief valve cable. Remove the spray lancer module.		
6	Remove the two screws securing the ESC's protective upper shell, then remove the upper shell.		

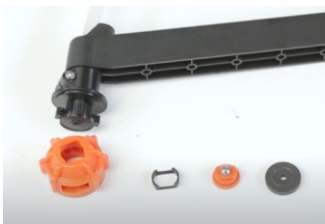
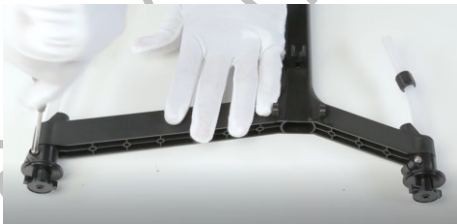
7	Remove the two screws securing the ESC's protective cover, then remove the cover.		
8	Remove the sealing rubber pads on the screws securing the ESC power cable connectors. Then remove the sealing rubber pad on the three-phase power supply cable.		
9	Remove the five screws securing the ESC power cable connectors and motor three-phase power supply cable.		
10	Remove the sealing rubbers as shown, and remove the three-phase power supply cable's waterproof rubber ring as shown.		
11	Remove the two screws securing the ESC signal cable connector, then disconnect the ESC signal cable.		
12	Remove the two screws securing the LED board's signal cable connector, then remove the ESC module.		

13	Remove the LED board.		
14	Remove the two screws securing the LED board's fixing piece, then remove the fixing piece.		
15	Remove the four screws securing the motor, then remove the motor.		
16	Remove the four motor pads.		
17	Remove the two screws securing the motor base, then remove the motor base.		
18	Remove the cable rubber and pull out the cable as shown.		Do not damage the cable.

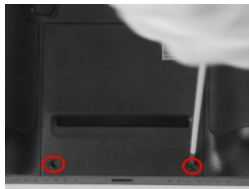
19	Remove the screws securing the hoses' fixing pieces, then remove the fixing pieces.		
20	Remove the four screws securing the thread sleeves. Then remove the thread sleeve A and thread sleeve B.		
21	Remove the damping rubbers.		

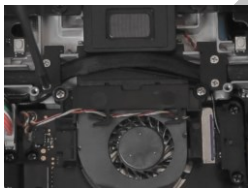
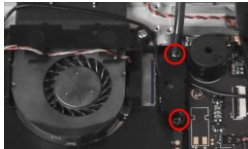
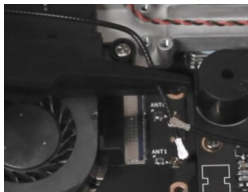
3.3.4 Disassembling the Spray Lance Module

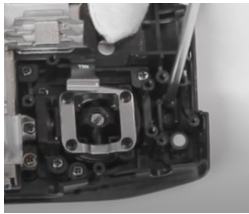
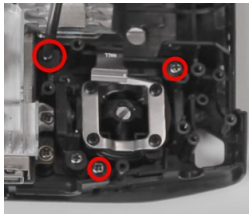
No.	Disassembly and Assembly	Picture	Notice
1	Remove the four screws securing the spray lance's fixing piece, then remove the fixing piece.		
2	Loosen the quick-release nut and remove all the white nylon sleeves on the electromagnetic relief valve's three-way valve.		

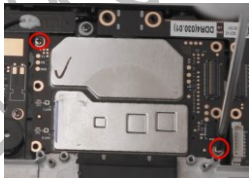
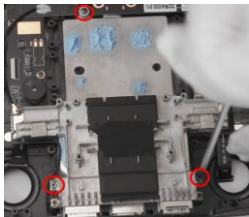
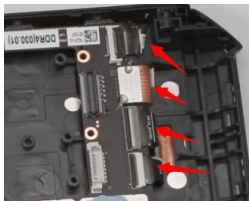
3	Remove the electromagnetic relief valve.		
4	Remove the two sprinkler modules.		
5	Then remove the sealing ring, nozzle, locating ring, and buckle.		
6	Remove the screws securing the two sprinkler bodies as shown, then separate the sprinkler body, white hose, and extended fixing rod.		
7	Loosen the quick-release nut and remove the white hose, sprinkler body, and nut.		

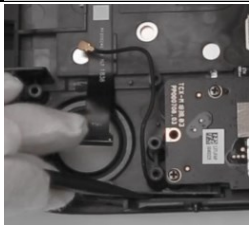
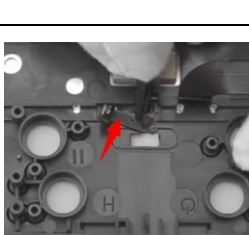
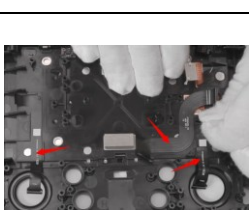
3.3.5 Disassembling the Remote Controller

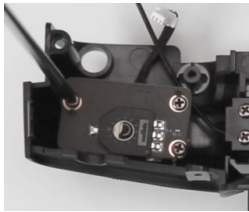
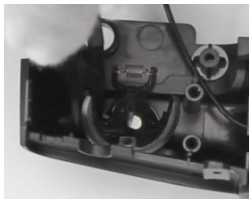
No.	Disassembly and Assembly	Picture	Notice
1	Remove the two screws securing the rear cover, and then remove the rear cover.		

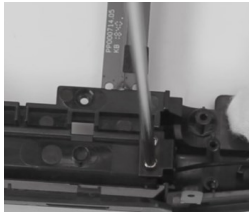
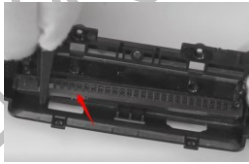
2	Remove the left and right rubbers.		
3	Remove the eight screws securing the lower shell, and then remove the lower shell.		
4	Remove the four screws securing the two battery cable mounting pieces, and then remove the cable mounting pieces.		
5	Pull out the battery plug and remove the battery.		
6	Remove the two screws securing the image transmission antenna mounting piece, and then remove the mounting piece.		
7	Disconnect the image transmission antenna connector.		
8	Disconnect the left and right dial flat cables.		
9	Remove the front cover's decorative piece.		

10	Remove the eight screws securing the front cover module, and then remove the front cover module.		
11	Remove the Focus/Shutter button, Record button, and spring.		
12	Remove the sticker of the flat cable which connects the main board and right Hall sensor board, and then remove the flat cable.		
13	Remove the four screws securing the right Hall sensor board, and then remove the right Hall sensor board.		
14	Remove the two screws securing the Focus/Shutter button board, and then remove the Focus/Shutter button board.		
15	Remove the screw securing the Confirm/Custom button C3, and then remove the Confirm/Custom button C3.		
16	Remove the three screws securing the right control stick, and then remove the right control stick. In the same way, remove the left Hall sensor board, record button board, Return/Function button board, and left control stick.		
17	Disconnect the fan flat cable and remove the two screws securing the fan module, and then remove the fan module.		

18	Remove the decorative piece on the SDR board.		
19	Remove the two screws securing the main board mounting piece (on the SDR board), and then remove the mounting piece and flat cable on the SDR board.		
20	Remove the flat cable from the main board to the left Hall sensor board, heat the UV glue at both ends of the cable connector with a hot air gun, clean the UV glue and unplug the speaker cable (hot air gun: 80°C, 2min), and then remove the speaker.		
21	Remove the two screws securing the SDR board and then remove the SDR board.		
22	Remove the three screws securing the heat sink and then remove the heat sink.		
23	Disconnect the Wi-Fi antenna 1 and Wi-Fi antenna 2 feeder, then remove Wi-Fi antenna 2.		
24	Remove the flat cable from the main board to the right Hall sensor board, flat cable from the main board and combination button board, screen flat cable (heat the UV glue at both ends of the cable connector with a hot air gun, hot air gun: 80°C, 2min), and flat cable from main board to mic board.		

25	Remove the four screws securing the main board, and then remove the main board.		
26	Remove the Wi-Fi antenna 1.		
27	Unscrew the 4 screws securing the combination button board, and remove the photosensitive board cable and cable connecting the main board and combination button board.		
28	Remove the silicone rubber sleeves of the power button, RTH button, and Pause button.		
29	Remove the power button, RTH button, Pause button, 5D button, and flight mode switch.		
30	Remove the photosensitive board.		
31	Disconnect the flat cable which connects the main board and right Hall sensor board, flat cable which connects the main board and combination button board, and flat cable which connects the main board and left Hall sensor board.		

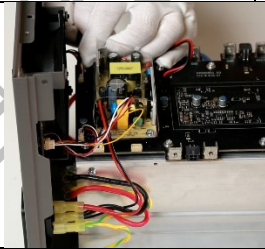
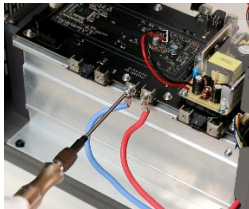
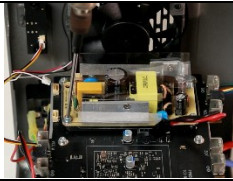
32	Remove the mic board.		
33	Remove the two pieces of thermal foam.		
34	Remove the screen module (put it in the oven, 70°C, 10min).		During repair, replace the upper shell module (with screen) directly.
35	Unscrew the two screws fixing the light guide beam on the front cover decoration piece, and then remove the light guide beam.		
36	Remove the motor on the lower shell module.		
37	Remove the two pieces of battery EVA foam.		
38	Remove the three screws securing the right dial board on the front cover, and then remove the right dial board.		
39	Remove the right dial module (aircraft control switch dial). In the same way, remove the left dial board and left dial (spray rate dial).		

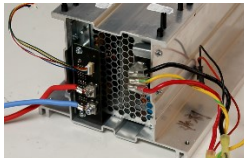
40	Use a hot air gun to blow on the connection part between the antenna sleeve and the antenna base, and remove the antenna sleeve by shaking it left and right with your hand (note that the hot air gun blows toward the sleeve side to avoid damaging the antenna base).		During repair, replace the antenna sleeve.
41	Remove the four screws securing the left and right antenna axes, and then remove the axes and image transmission antenna.		
42	Remove the antenna base.		
43	Remove the air outlet decorative piece and dustproof mesh.		

3.3.6 Disassembling the Charger

No.	Disassembly and Assembly	Picture	Notice
1	Check whether the anti-dismantle sticker on the charger is removed. If yes, no warranty repair service will be provided. When mounting the screws, the screw glue is required.		The anti-dismantle sticker needs to be attached after installation.
2	Remove the four charger plugs.		
3	Remove the six screws securing the two charger side boards C on the charger side board A module. Remove the charger side boards C.		
4	Remove the two charger battery charging cables.		There is a cable clamp on the communication cable, so it cannot be removed with too much force.

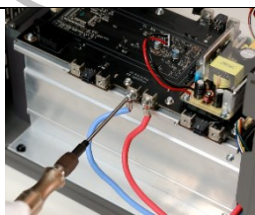
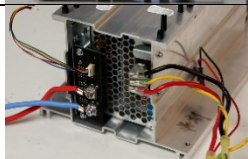
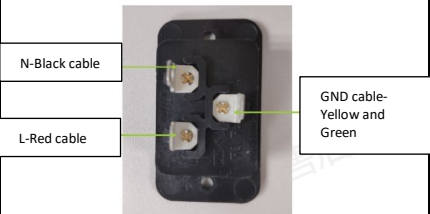
			Note that the red cable is attached to the left small screw hole, and the blue cable is on the right larger hole. The cables must be attached to the correct positive and negative poles, and the screws on the left and right are different. Torque for installation: 5.0-5.5kgf.cm
5	Remove the two charger side boards C on the charger side board A module, and then remove the two charging cables in the same way.		
6	Remove the four screws on the charger rubber pads, and then remove the rubber pads.		Do not use too much force during installation. Otherwise, the pads may be damaged by the screws.
7	Remove the four screws securing the charger upper cover module.		
8	Remove the upper cover and disconnect the communication cable on the charger button board.		Slowly remove the upper cover to avoid damaging the communication cable.
8.1	Disconnect the power cable and use the long nose pliers to remove the power button.		Do not lose the waterproof rubber ring. The whole switch part is a material. When installing, pay attention not to install the switch in the opposite direction. Ensure that the circular silk print faces downwards.
8.2	Remove the four screws securing the LED board, then remove the LED board and shielding foam.		
8.3	Remove the charger silicone rubber stopper and button cap.		

8.4	Remove the two screws' shielding pieces on the handle.		
8.5	Remove the two screws and nuts securing the handle.		
9	Remove the four screws securing the charger fixing plate, and then remove the fixing plate.		
10	Remove the two charger side board A modules.		When installing, arrange the input cables to prevent them from being crushed during installation.
11	Disconnect the adaptive board's communication cable, fan's cable, four-core cable connecting the USB Type-C to main board, AC power cable, auxiliary power board cable, and button board's communication cable.		
12	Remove the screws securing the AC power red cable (positive) and AC power blue cable (negative).		During installation, ensure that the red cable is attached to the small screw hole (positive pole) while the blue cable is attached to the large screw hole (negative pole). Torque: 5.0-5.5 kgf.cm
13	Remove the four screws securing the auxiliary power board, and then remove the board. Remove the charger auxiliary power vacuum thermoform cover and main board anti-dust vacuum thermoform cover as shown.		
14	Remove the eight screws securing the charger management main board PCBA (with firmware), and then remove the PCBA.		
15	Remove the eight cooling silicone pads.		If the cooling pad is replaced, thermal oil must be applied.

16	Remove the four screws securing the 4850 power module. Then remove the GND cable and 4850 power module.		The GND cable (yellow, green cable) needs to be attached to the middle connector on the 4850 power module.
17	Disconnect the four-core cable connecting the USB Type-C and main board. Remove the two screws securing the USB update adaptive board. Remove the adaptive board.		
18	Remove the two screws securing the AC power plug, and then remove the plug.		Pay attention to the installation direction of the power plug.
19	Remove the four screws securing the fan's metal cover, and then remove the cover.		If only the fan is damaged, there is no need to remove the upper cover.
20	Remove the four screws securing the fan and then remove the fan. Then remove the foam as shown.		Pay attention to the installation direction. Make sure that the sticker with the silk print faces outward, and that the cable is near the yellow sticker.
21	Disassembling the 4850 Power Module		
21.1	Disconnect the AC power input cables (negative and positive), AC power input cable GND, AC power output blue cable (negative), AC power output red cable (positive), and adaptive board's communication cable.		Ensure that the fire wire, zero line, and earth wire are attached correctly. The red and blue cables cannot be connected incorrectly. Torque: $5.5 \pm 0.5 \text{ kgf.cm}$
21.2	Remove the two screws securing the charger 3KW bracket-02, and then remove the 3KW bracket-02.		
21.3	Remove the 3-pin adapter.		
21.4	Remove the two screws securing the MG-2 charger power adaptive board, and then remove the board.		

21.5	Remove the two screws securing the charger 3KW bracket-01, and then remove the 3KW bracket-01.		
21.6	Remove the four screws securing the power module bracket B, and then remove the bracket and power board. Separate the charger power module aluminum bracket and 4850 power module as shown.		
21.7	Remove the four screws securing the fan module, clean off the white glue, disconnect the cable, and then remove the fan.		During assembly, the white glue needs to be applied.

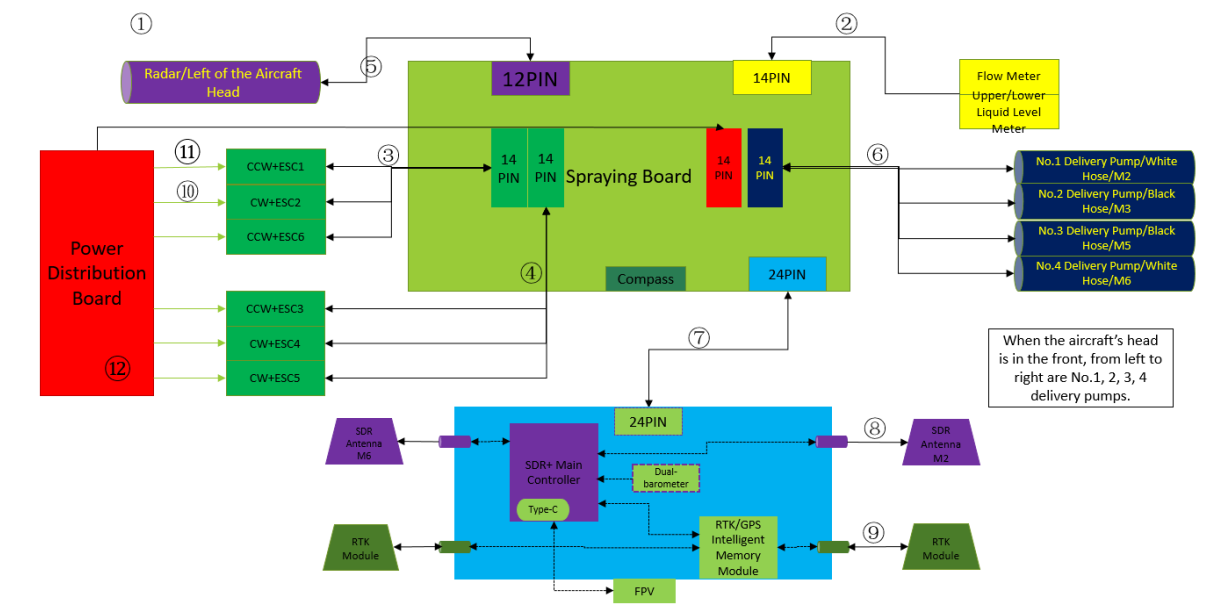
3.3.7 Charger Assembly Notices

1	Ensure that the torque is 5.0-5.5Kgf.cm when connecting the red and blue power cables.		
2	Ensure that the torque is 5.0-5.5Kgf.cm when connecting the battery charging cable.		
3	The live cable has a mark of "L", and the zero cable has a mark of "N". The other is the ground cable connected to the yellow and green cable, and the end is connected to the 4850 module of the charger for safety. Please make sure that the ground wire (yellow, green cable) is connected to the connector on the middle of the module, as shown in the figure.	 	

4 Aircraft Principle

4.1 Connection Diagram and Board Functions

4.1.1 Connection Diagram



The aircraft consists of the following six modules: aerial-electronics system, radar system, spray system, propulsion system, vision positioning system, and image transmission system. These six modules contain a total of 18 sub-modules.

1. Aerial-Electronics System Module

The aerial-electronics board is the core module controlling the flight controller. It integrates the flight algorithm, image transmission, and flight controlling functions into one module. It manages the aircraft's resources.

2. Omnidirectional Radar

The omnidirectional radar is a sensor system for obstacle avoidance and navigation of the aircraft. It is responsible for detecting the external environment and preventing the aircraft from hurting people and crashing.

3. Spraying Board

The spraying board is the control system of the aircraft's core spraying function. It controls the pump and spraying volume.

4. Power Distribution Board

The power distribution board is a power transfer module that distributes power to the six motors.

5. Propulsion ESC Board

The ESC is responsible for converting the throttle information into the motor speed information, controlling

the motor rotation, and feeding back the motor speed and overcurrent protection.

Input Voltage	14S Maximum: 60 V
Input Power	58 V, 40 A
Maximum Current	120 A
Protection Function	Self-check, blocking protection, and short circuit protection
Protection Rating	IP67
Assembly Technology	The ESC shell is assembled and replaced independently. The motor and power cables are fixed with screws, and the signal and LED cables are attached with waterproof connectors.

6. LED Board

The LED board indicates the aircraft's current working condition. Users can know the aircraft's current working condition by the color of the LEDs.

7. FPV Board

The FPV board transmits the camera information to the aerial-electronics board, moves the camera, and provides night illumination.

8. Barometer Board

The barometer board mainly measures the current air pressure signal and transmits the air pressure value to the aerial-electronics board.

9. Delivery Pump ESC Board

The delivery pump ESC board is the power system of the delivery pump. It controls the delivery pump motor rotation based on the signal of the spray board.

10. Liquid Level Meter

The liquid level meter tells the spraying board the dose information and reminds the user when it is necessary to add liquid.

11. Flow Meter

The flow meter is essential for high-precision spraying. It detects the pump's spraying speed and reports the information to the aerial-electronics board. In this way, the pump's spraying accuracy can be controlled.

12. RTK Antenna

The RTK module provides reliable information such as the position, direction, and speed of the aircraft for the flight controller system to ensure the stability of the aircraft's navigation as well as the efficiency and accuracy of operations. The RTK module contains functional circuits such as RTK, GPS, data protector and related power management. The RTK can track GPS L1/L2, GLONASS F1/F2, BDS B1/B2, and Galileo E1/E5b frequency bands at the same time. The GPS module is used with the main antenna of the RTK, accepting GPS L1 and GLONASS F1 frequency band signals.

13. Active Antenna Board

The active antenna board amplifies and adjusts the GPS weak signal to make the RTK/GPS board receive better satellite signals.

14. Motor

The motors work with the propellers to blow wind downwards and make the aircraft fly upwards. It also helps with spraying, so that the pesticide can penetrate more thoroughly.

15. Delivery Pump

The delivery pump is the executor of the aircraft spraying system, which sprays pesticides out by rotating.

16. FPV

FPV, first person vision, is a camera module that converts optical signals into electrical signals and transmits them to the remote controller through image transmission.

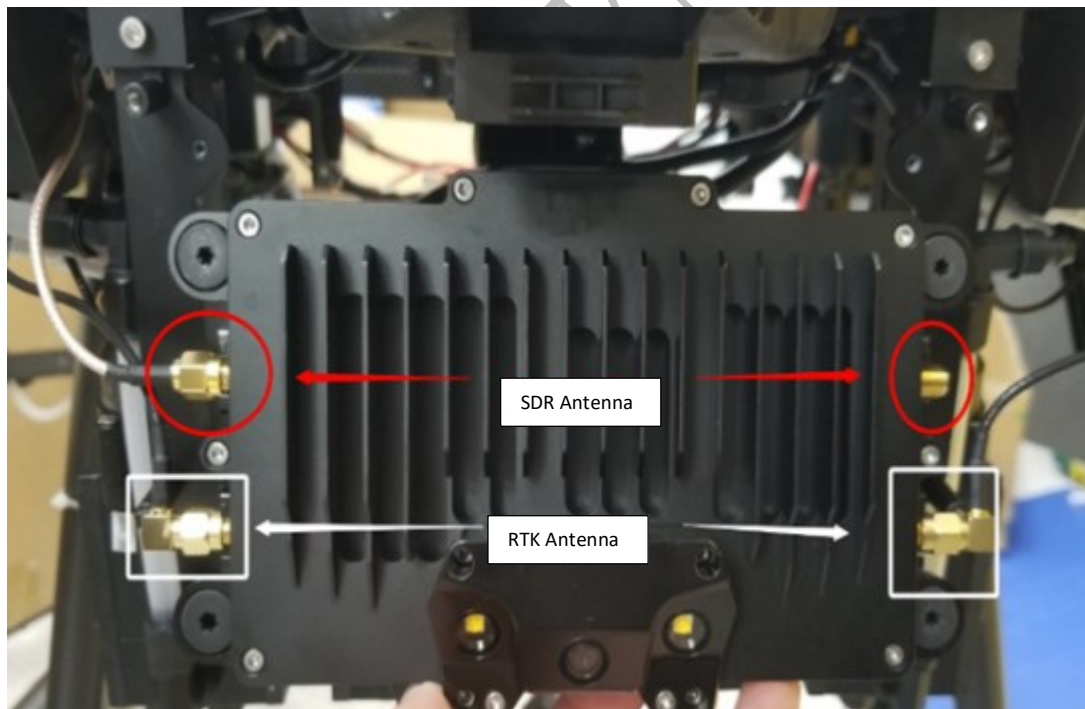
17. Battery

The battery is the energy source for flight and spraying, and is an energy storage equipment. The number of battery cells is 14S, and the capacity is 18000 mAh.

18. Electromagnetic Exhaust Valve

The electromagnetic exhaust valve has an automatic pressure relief function, which can automatically empty the air in the hose. When the delivery pump is turned on, the air is automatically exhausted. When the delivery pump is turned off, the electromagnetic exhaust valve will stop operating.

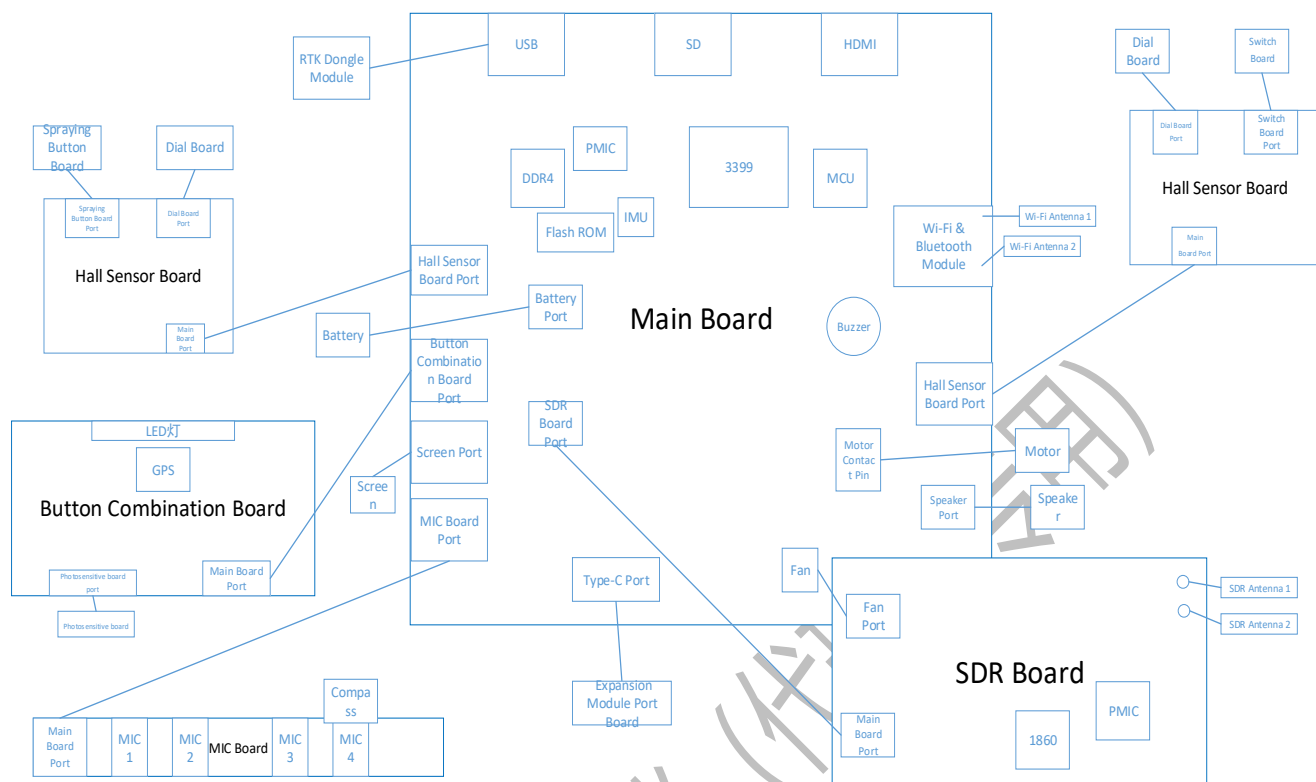
4.1.2 Diagram



No.	Material Number	Material Name	Function
1	YC.XC.XX000617	Signal Cable Connecting the Power Distribution Board and Spraying Board	Connects the power distribution board with the spraying board.

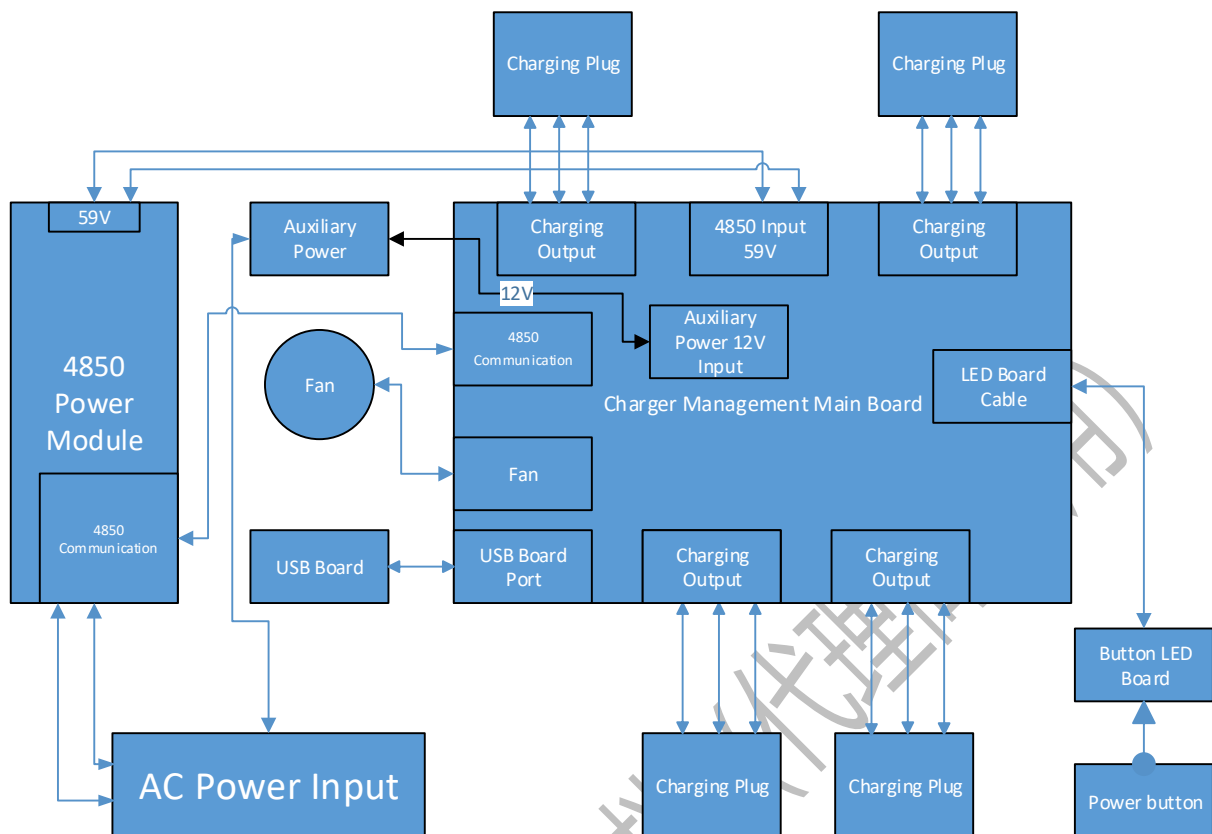
2	YC.XC.XX000518	Liquid Level Meter Cable	The spraying board connects the flow meter and liquid meter.
3	YC.XC.XX000513	Signal Cable A Connecting the Spraying Board and Propulsion ESC	Connects the No.1, 2, and 3 aircraft arms to the spraying board
4	YC.XC.XX000512	Signal Cable B Connecting the Spraying Board and Propulsion ESC	Connects the No.4, 5, and 6 aircraft arms to the spraying board
5	YC.XC.XX000510	Radar Power Cable	Connects the spraying board to the radar.
6	YC.XC.XX000511	Power Cable Connecting the Spraying Board and Delivery Pump	Connects the spraying board and four delivery pumps
7	YC.XC.XX000519	Aerial-electronics _Power Cable	Connects the aerial-electronics board to the spraying board
8	YC.DZ.AA000086	SDR Antenna	
9	YC.XC.TZ000067	RTK Antenna Feeder	Connects the RTK antenna and aerial-electronics module
10	YC.XC.DD000348	Power Distribution Board Power Cable (Negative) (M4/5/6)	Connects the No.4, 5, and 6 aircraft arms to the power distribution board
11	YC.XC.DD000349	Power Distribution Board Power Cable (Positive) (M1/2/3)	Connects the No.1, 2, and 3 aircraft arms to the power distribution board
12	YC.XC.DD000347	Power Distribution Board Power Cable (Positive) (M4/5/6)	Connects the No.4, 5, and 6 aircraft arms to the power distribution board
13	YC.XC.DD000350	Power Distribution Board Power Cable (Negative) (M1/2/3)	Connects the No.1, 2, and 3 aircraft arms to the power distribution board

4.2 Remote Controller Connection Diagram and Board Function

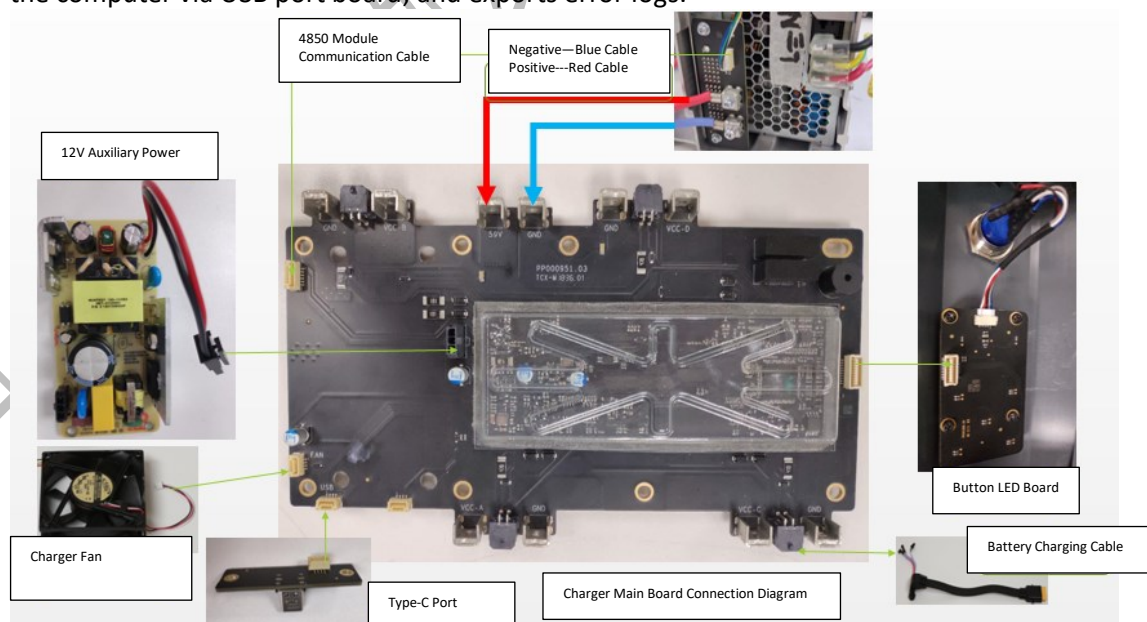


Board	Function
Main Board	Receive and process the information of each module of the remote controller
SDR Board	Process the image transmission signal of the aircraft
MIC Board	Has a microphone function
Hall Sensor Board	Receive dial and button information
Button Board	Report each button and photosensitive board information, including GPS
Photosensitive Board	Receive light intensity and feed it back to the main controller
Expansion Kit Port Board	Connects to the external battery and has a dongle function

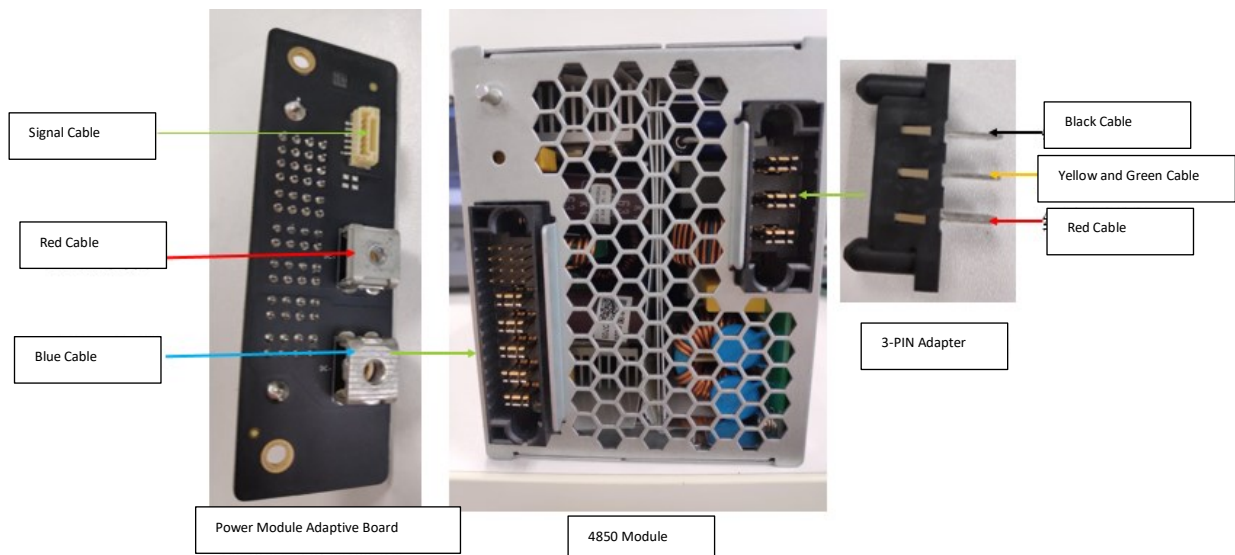
4.3 Charger Connection Diagram



1. Charger management main board: Communicates with the battery; manages and controls the four output ports for charger fast charging, slow charging, and determines which battery is charged first; controls the color of LEDs based on the charging situation and charger status; communicates with the computer via USB port board; and exports error logs.



2. 4850 power module: Transfers electric supply power to 59V DC power supply, and charges the battery and communicates with the charger management main board. Torque of red cable and blue cable: 5.0-5.5Kgf.cm



3. Auxiliary power source: Transfers electric supply power to the 12V DC power supply and supplies power to the charger management main board.
4. Battery charging port: Port connecting the charger management main board and battery, and provides charging voltage and communication. Torque for connecting the cable: 5.0-5.5 Kgf.cm
5. LED board: Connects the switch, and transfers the switch and main board; there are seven LEDs on the board, indicating the charging status of each individual port, warning, and fast/slow charging display.
6. Fan: Cool down the charger.
7. USB port adaptive board: Adapts the USB port to the main board.

5 Damage Assessment and Repair Guide

5.1 Aerial-Electronics System Module

5.1.1 Aerial-electronic system module: flight controller, SDR, barometer

Module	Malfunction	Analysis	Solution
Aerial-electronic system module (flight controller, SDR, barometer)	DJI Assistant 2 for MG cannot be connected	1. The firmware is lost.	Update the firmware again.
		2. The DJI Assistant 2 for MG version does not match.	Download the DJI Assistant 2 for MG of the corresponding version on the DJI official website.
		3. The computer drive installation fails.	Reinstall the drive or DJI Assistant 2 for MG or use a different computer.
		4. Poor connection	Replace the aerial-electronics module.
	Update failure	1. The firmware is lost.	Update the firmware again.
		2. There is an error with the network.	Ensure that the network is normal and update the firmware again.
		3. The aerial-electronics system is not completely connected.	Ensure that all modules are completely connected.
		4. Check the corresponding updating error code.	Check the code of the corresponding component.
	The aircraft hovers and drifts.	1. There is an error with the IMU calibration.	Perform the IMU (6-directional) calibration.
		2. When the satellite search is normal and the motor works normally, check whether the aerial-electronics system is abnormal (There is no obstruction or abnormal sounds, and the RPM value is normal).	Replace the aerial-electronics module.
	The aircraft cannot fly in a straight line.	1. There is an error with the compass calibration.	Re-calibrate the compass or replace the spraying module.
		2. When the compass calibration is normal, check whether the aerial-electronics module is abnormal.	Replace the aerial-electronics module.
	There is an error with the master remote controller data (microSD card).	Power off the aircraft for 5 minutes and then power it on to see if it has returned to normal.	Replace the aerial-electronics module.
	There is an error with the master remote controller data (IMU).	Perform the IMU (6-directional) calibration.	If the calibration fails, replace the aerial-electronics module.
	Water damage	The liquid contact indicator has turned red or the boards are corroded.	Replace the aerial-electronics module.
	Linking failure	1. Check whether the remote controller matches with the aircraft firmware.	Update the firmware to the same version or latest version.
		2. Check whether the power distribution board port is abnormal.	Replace the power distribution board module.

	The transmission distance is short or there is no image transmission. Or, the screen is frozen.	1. Check whether the remote controller matches with the aircraft firmware. 2. Replace the remote controller or the aerial-electronics module for cross testing and check which component is damaged.	Update the aircraft and remote controller to the latest version. Replace the remote controller first, and if the issue persists, replace the aerial-electronics module.
	(N/A) Country code display (N/A)	The country code is lost.	Return to the factory and refresh the country code.
Barometer	The aircraft height determination is incorrect or abnormal.	Replace the aerial-electronics module and test again.	Replace the aerial-electronics module.

5.1.2 Aerial-electronic system module: FPV module

Module	Malfunction	Analysis	Solution
Aerial-electronic system module: FPV and spotlight	The spotlight does not light up. / There is no image transmission. / The image transmission screen freezes.	Check whether the remote controller firmware matches with the aircraft firmware.	Update the firmware to the same version or latest version.
		Replace the remote controller and test again.	Replace the remote controller.
		The DJI MG app version is too old.	Update the DJI MG app.
		The spotlight function is not enabled in the app.	Enable the spotlight function in the app.
		Replace the aerial-electronics board and test again.	Replace the aerial-electronics module.

5.1.3 Aerial-electronic system module: RTK/GPS/Data protector module

Module	Malfunction	Analysis	Solution
Aerial-electronic system module: RTK/GPS/Data protector Module	The GNSS satellite search fails or the satellites are lost.	1. There is environmental disturbance or the firmware is inconsistent.	Try in a different environment or update the firmware.
		2. The RTK antenna contact is poor.	Reinsert the RTK antenna.
		3. Replace the RTK antenna to cross test and check whether it is damaged.	Replace the RTK antenna.
	The aircraft will drift or frequently switch the flight mode to Atti mode.	1. There is environmental disturbance or the firmware is inconsistent.	Try in a different environment or update the firmware.
		2. The RTK antenna contact is poor.	Reinsert the RTK antenna.
		3. Replace the RTK antenna to cross test and check whether it is damaged.	Replace the RTK antenna.
		Replace the spraying board to cross test (test the spraying board compass).	Replace the spraying board.
	There is an error report for the RTK angle.	Ensure that the left and right antennas are connected correctly.	Check the antenna connection sequence.
	The number of RTK satellites is not stable / Switch the	There is environmental disturbance or the firmware is inconsistent.	Try in a different environment or update the firmware.

	RTK to the GPS mode.	2. The RTK antenna contact is poor.	Reinsert the RTK antenna.
		3. Replace the RTK antenna to cross test and check whether it is damaged.	Replace the RTK antenna.
		Replace the aerial-electronics module and test again.	Replace the aerial-electronics module.

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5.2 Spraying module

Module	Malfunction	Analysis	Solution
Spraying module	The battery communication cannot be read.	1. The battery signal cable of the power distribution board is damaged.	更换分电板模块 Replace the power distribution board.
		2. The cable that connects the spraying module and the aerial-electronics module is damaged.	Replace the cable that connects the spraying module and the aerial-electronics module.
		3. Replace it with a normal spraying module for testing.	Replace the spraying module
	Compass calibration fails. / There is no compass value.	1. There is environmental disturbance.	Calibrate in a different environment.
		2. The compass module is damaged.	Replace the spraying module
	Water damage	The liquid contact indicator has turned red or the board pieces are corroded.	Replace the spraying module

5.3 Delivery pump

Module	Malfunction	Analysis	Solution
Delivery pump	There is air inside the delivery pump/ the delivery pump is powerless or cannot work.	The delivery pump is not calibrated.	Calibrate the delivery pump and flow meter
		2. Replace the delivery pump and hose for cross testing.	Ensure that the hose is connected well and the nozzle is not leaking.
		3. Check the cable which connects the delivery pump and spraying module is connected well.	Replace the cable.
		4. Check whether the delivery pump is damaged.	Replace the delivery pump.
		Check whether the spraying module is damaged.	Replace the spraying module.
	There is water leakage.	The membrane or the structure is damaged.	Replace the delivery pump.

5.4 Liquid Level meter

Module	Malfunction	Analysis	Solution
Liquid level meter	The spray tank pesticide reading is incorrect. / The spray tank pesticide cannot be read.	1. The liquid level meter is not mounted properly or is obstructed and does not change based on the liquid level.	Re-install the liquid level meter.
		2. The liquid level meter cable is not attached properly.	Attach the liquid level meter cable again.
		3. The liquid level meter cable or the liquid level meter is damaged.	Replace the liquid level meter and liquid level meter cable.

5.5 Radar

Module	Malfunction	Analysis	Solution
Radar	The height cannot be detected. / The obstacle avoidance fails.	1. Check whether the radar switch is enabled in the app.	Enable the radar switch.
		2. Check whether there are objects inside.	Clean the interfering objects.
		3. Check whether the radar firmware is updated to the latest version.	Update the firmware.
		4. Check whether the radar bracket is misshapen.	Replace the radar bracket.
		5. Check whether the radar cable is damaged.	Replace the radar cable.
		Replace the radar and test again.	Replace the radar.
	Water damage	The internal board is corroded and damaged due to oxidation.	Replace the radar.

5.6 Flow Meter

Module	Malfunction	Analysis	Solution
Flow meter	The flow meter information cannot be read. / The flow information is incorrect.	1. Check whether the cable is damaged or has poor contact.	Reinsert or replace the cable.
		2. The flow meter is not calibrated.	Calibrate the flow meter and delivery pump.
		3. Check whether the flow meter is damaged.	Replace the flow meter.
		4. Check whether the spraying module is damaged.	Replace the spraying board.

5.7 Electromagnetic Relief Valve

Module	Malfunction	Analysis	Solution
Electromagnetic relief valve	It is leaking or does not work.	1. The connection is poor	Remove and attach the power cable of the electromagnetic relief valve module.
		2. The electromagnetic relief valve is damaged.	Replace the electromagnetic relief valve module.

5.8 Power Distribution Board

Module	Malfunction	Analysis	Solution
	The battery communication is	1. Check whether the power cable has short-circuited or disconnected.	Replace the power cable.

Power distribution board	poor. / Unable to power on.		
		2. Check whether the cable is damaged.	Replace the battery signal cable.
		3. Replace with another normal battery for cross testing.	Replace the battery.
		4. Replace the spraying board for cross testing.	Replace the spraying board.
		5. Replace the aerial-electronics module for cross testing.	Replace the aerial-electronics module.

5.9 Motor/ESC/LED

Module	Malfunction	Analysis	Solution
Motor/ESC	All motors cannot be enabled.	1. Check whether the cable is properly connected.	Replace the connection cable.
		2. Check whether there is an abnormal prompt in the app.	Check the corresponding component based on the prompt.
		3. Check whether the spraying board is damaged.	Replace the spraying module
		4. Check whether the aerial-electronics board is damaged.	Replace the aerial-electronics module.
	No. 1, 2, and 3 motors report a warning that it cannot be enabled.	1. Check whether the cable port is damaged.	Reinsert or replace the cable.
		2. Check whether the spraying board is damaged.	Replace the spraying board.
	No. 4, 5, and 6 motors report a warning that it cannot be enabled.	1. Check whether the cable port is damaged.	Reinsert or replace the cable.
		2. Check whether the spraying board is damaged.	Replace the spraying board.
	The single motor cannot be enabled.	1. Check whether the ESC firmware is normal.	Update the firmware.
		2. Check whether the corresponding cable is damaged.	Replace the cable.
		3. Check whether the three-phase cable of the motor is free of glue melting or breaking. Or, check whether the riveting has broken.	Replace the motor.
		4. Check whether the axle gear screw of the motor rotor is loose or broken.	Replace the motor.
	The motor makes abnormal sounds or cannot rotate smoothly.	The motor axis is misshapen or worn.	Replace the motor.

LED	The ESC light does not turn on or does not have a color.	1. The ESC board is damaged.	Replace the ESC board.
		2. The LED is damaged.	Replace the LED.

5.10 Charger & Battery

Module	Issue	Cause Analysis	Solution
Charger	The charger does not power on (the LED does not light up and the fan does not rotate).	The auxiliary power is damaged, so there is no 12 V output for the charger management main board.	Replace the auxiliary power.
		The charger management main board is damaged.	Replace the charger management main board.
		The switch is damaged.	Replace the switch.
		The cable connecting the button board and the charger management main board is damaged.	Replace the corresponding cable.
		The AC power cable is damaged, or the negative and positive cables are loose.	Replace or reinsert the AC power cable, negative cable, and positive cable.
	The charger can be powered on, but the ports cannot be used to charge.	The 4850 power module is damaged.	Replace the 4850 power module.
		The charger management main board is damaged.	Replace the charger management main board.
	One port cannot be used to charge.	The corresponding charging cable is not connected well or damaged.	Perform a visual inspection and replace the charging cable.
		The corresponding charging cable is not connected well or damaged.	Perform a visual inspection and replace the charging plug.
		The charger management main board is damaged.	Replace the charger management main board.
	The power button LED lights up, but the fan does not rotate.	The fan is damaged.	Replace the fan.
		The corresponding charging cable is not connected well or damaged.	Replace the charger management main board.
	Insert the battery and the corresponding port's LED blinks yellow continuously.	The corresponding charging cable is damaged or not connected well.	Perform a visual inspection and replace the charging cable.
		The corresponding charging plug is damaged or not connected well.	Perform a visual inspection and replace the charging plug.
	Insert the battery and the corresponding port's LED blinks	There is an error with the battery.	Replace the battery.

	yellow twice or four times.		
	Insert the battery and the corresponding port's LED blinks continuously (after the battery is inserted for about 10 seconds).	The battery is not inserted well or a non-official battery is used.	Replace the battery.
	Insert the battery and the corresponding port's LED blinks yellow twice.	The port is short-circuited.	Check whether the charging plug or charging cable is short-circuited.
	Insert the battery and the corresponding port's LED does not light up and the warning indicator is solid yellow.	The 4850 power module is damaged.	Replace the 4850 power module.
		The charger management main board is damaged.	Replace the charger management main board is damaged.
	Insert the battery and the corresponding port's LED does not light up and the warning indicator blinks twice.	The fan of the 4850 power module is damaged.	Replace the fan and 4850 power module.
	Insert the battery and the corresponding port's LED does not light up and the warning indicator blinks three times.	The charger's fan is damaged.	Replace the charger's fan.
	Insert the battery and the corresponding port's LED does not light up, the warning indicator blinks once, and the charger makes "DDD" sounds.	The fan of the 4850 power module is damaged.	Replace the 4850 power module.

	Insert the battery and the corresponding port's LED does not light up, the warning indicator lights up, and the charger makes "DD.DD.DD" sounds.	The charger management main board is damaged.	Replace the charger management main board is damaged.
	The USB port cannot be used to connect and update.	The USB adaptive board is damaged.	Replace the USB adaptive board.
		The charger management main board is damaged.	Replace the charger management main board is damaged.
	The fast charging and slow charging modes cannot be switched.	The LED board is damaged.	Replace the LED board.
		The charger management main board is damaged.	Replace the charger management main board is damaged.

Module	Malfunction	Analysis	Solution
Battery	The battery cannot be turned on.	The battery management main board is damaged.	Return to the factory for repair.
	The battery communication fails and LED1/2 blink quickly.	The battery port board is damaged.	Return to the factory for repair.
		The battery management main board is damaged.	Return to the factory for repair.
	The LED1 and LED4 blink quickly.	The battery cell is damaged.	Return to the factory for repair.
		The battery management main board is damaged.	Return to the factory for repair.
	The battery can be powered on or off normally, but there is no output and it cannot be charged.	The fuse is scorched.	Return to the factory for repair.
	Insufficient battery life	The battery capacity is virtually high.	Export the QR code data and send it to DJI after-sales for analysis.

5.11 Remote Controller

Module	Malfunction	Analysis	Solution
Remote controller	Unable to charge	1. Check whether the remote controller's built-in battery is damaged.	Replace the remote controller's built-in battery.

	Unable to power on	1. Check the external battery's battery level.	Replace the expansion module's port board.
		2. Check whether the battery cable is loose.	Tighten or replace the cable.
		3. Check the built-in battery.	Replace the built-in battery.
		4. Check the main board module.	Replace the main board
	DJI Assistant 2 for MG cannot be connected.	1. The DJI Assistant 2 for MG version is not matched.	Download the corresponding version of DJI Assistant 2 for MG on the DJI official website.
		2. The computer drive installation fails.	Reinstall DJI Assistant 2 for MG.
		3. Replace a normal main board for testing.	Replace the main board
	Linking failure	1. Check whether the remote controller firmware matches with the Air System.	Update remote controller firmware.
		2. Replace the SDR board for testing.	Replace the SDR board.
		3. Replace a normal main board for testing.	Replace the main board.
	The aircraft signal is lost in the specified range.	1. Check whether the antenna is damaged.	Replace the antenna.
		2. Perform the remote controller SDR coupling test.	Replace the SDR board.
		3. Perform the remote controller SDR noise level test.	Replace the SDR board.
		4. Replace the SDR board for testing.	Replace the SDR board.
		5. Replace a normal main board for testing.	Replace the main board.
	The screen is abnormal.	Check whether the screen or the coaxial cable is damaged.	Replace the screen.
	The remote controller satellite search is abnormal.	Replace a normal main board for testing.	Replace the button combination board.
	The Wi-Fi cannot be connected.	1. Check whether the Wi-Fi antenna is damaged or is not connected.	Tighten or replace the Wi-Fi antenna.
		2. Replace a normal main board for testing.	Replace the remote controller main board.
	The dial or button is malfunctioning.	Check whether the adapter plate is damaged.	Replace the Hall sensor board.
Expansion module	The battery cannot be mounted.	The type-C port of the remote controller main board is damaged.	Replace the remote controller main board.
		The type-C port of the expansion module is damaged.	Replace the expansion module's port board.
	The 4G dongle cannot be connected.	The type-C port of the remote controller main board is damaged.	Replace the main board
		The type-C port of the expansion module is damaged.	Replace the expansion module's port board.
RTK Dongle	The RTK satellite search is abnormal.	1. Check the RTK port.	Tighten or replace the RTK board.
		2. Check the remote controller main board port.	Replace the remote controller main board.

6 Test Items

6.1 Test

Mod ule	N o.	Test	Aircraft 飞行器																	
			Aerial- Electro nics Modul e	Spray ing Mod ule	Deliv ery Pum p *4	Liq uid Lev el Me ter *2	Rad ar	Flo w Me ter	Electroma gnetic Valve	Power Distri bution Board	Mo tor *6	ES C * 6	LE D *6	RTK Ante nna *2	SDR Ante nna *2	Aerial- Electro nics Power Cable	Spray ing Boar d Powe r Signa l Cable	Liq uid Lev el Me ter Cable	线 Rad ar Po wer Signa l Cable	ESC Comp osite Cable* 2
Aircr aft	1	Positi ve and negati ve & aircra ft body short circuit test	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	2	Firmw are packa ge updat e	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
	3	Flight testin g	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√

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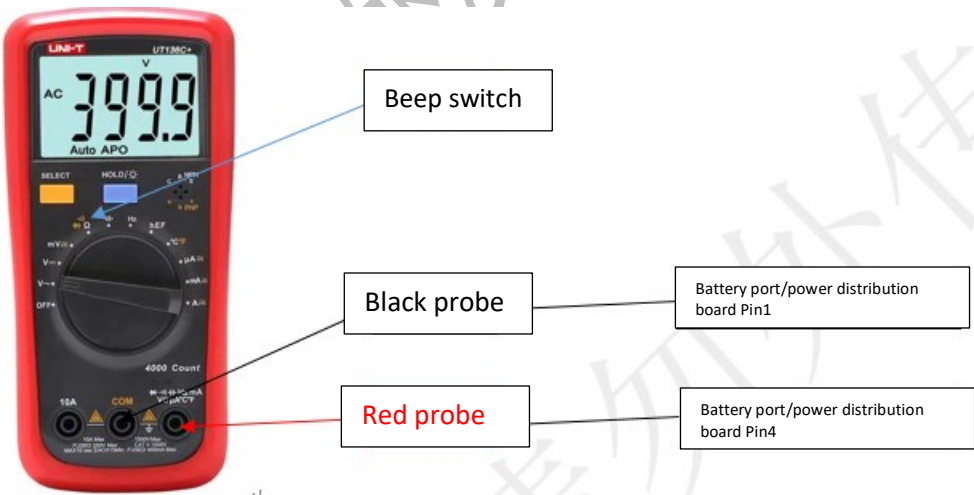
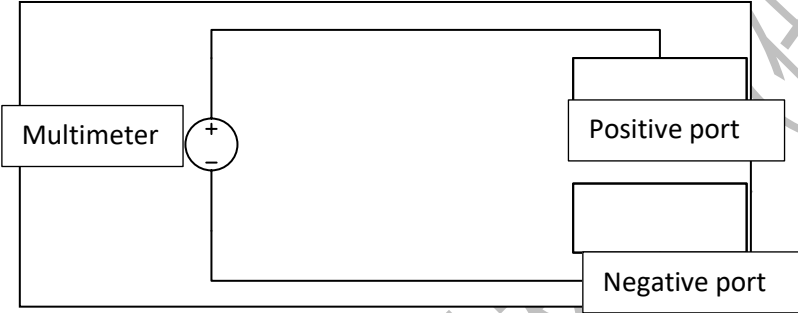
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6.2 Test Tool

Name	Model	Picture	Usage
Multimeter	Fluke		Short-circuit test

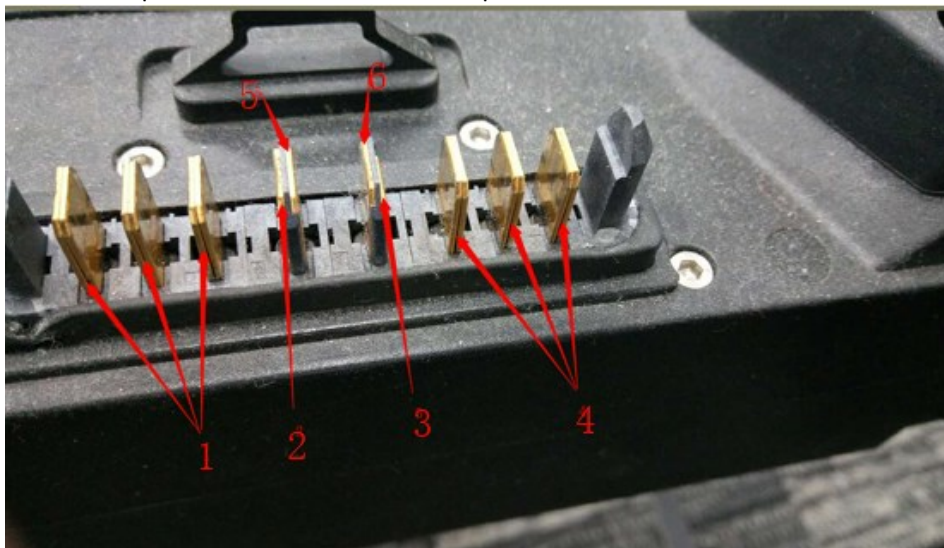
6.3 Test Connection Diagram

Short-circuit test

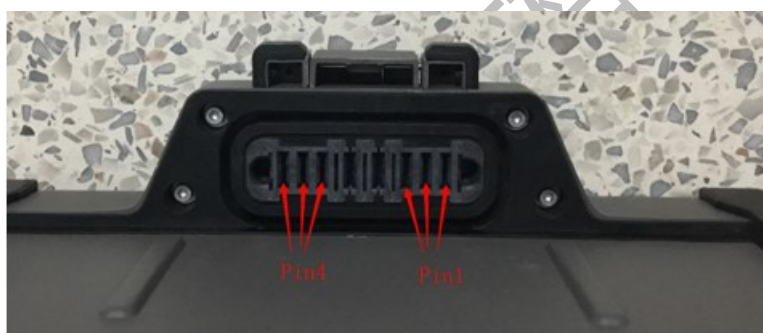


- 1. Toggle the switch to the beep switch.
- 2. Plug the black probe into COM and the red probe into VΩ, and connect the two probes to the two terminals (battery port/power distribution board Pin1 and Pin4) as shown above (Pin1 can also be measured by the red probe, and Pin4 can be measured by the black probe).
- 3. We recommend measuring the terminals three times. If the multimeter beeps during one of the measurements, the aircraft or battery internal power supply system is short-circuited. In this case, the powering on test is not allowed.

4. If the aircraft internal power supply system is short-circuited, check whether the propulsion cable is short-circuited. Check if the aircraft is water damaged, if a short circuit occurred due to hitting a high-voltage line, or if the ESC power cable is broken.
5. If the battery internal power supply system is short-circuited, stop using it and contact the DJI after-sales department.
6. The aircraft power distribution board and ports Pin1 and Pin4 are shown below:



Pin1:14S- Pin2: VCC5V Pin3:VCC5V_GND Pin4:14S+ Pin5: 485A Pin6:485B



Pin1:14S- Pin4:14S+

7 Product Comparison

Specifications	T20	T16
Total Weight (Excluding battery)	21.1 kg	18.5 kg
Standard Takeoff Weight	42.6 kg	39.5 kg
Max Takeoff Weight	47.5 kg (at sea level)	40.5 kg (at sea level)
Max Thrust-Weight Ratio	1.70 (with a takeoff weight of 47.5 kg)	2.05 (with a takeoff weight of 39.5 kg)
Hovering Accuracy (With a strong GNSS signal)	D-RTK enabled: Horizontal: ± 10 cm, Vertical: ± 10 cm D-RTK disabled: Horizontal: ± 0.6 m, Vertical: ± 0.3 m (Radar module enabled: ± 0.1 m)	D-RTK enabled: Horizontal: ± 10 cm, Vertical: ± 10 cm D-RTK disabled: Horizontal: ± 0.6 m, Vertical: ± 0.3 m (Radar module enabled: ± 0.1 m)
RTK/ GNSS Operating Frequency	RTK: GPS L1/L2, GLONASS F1/F2, BeiDou B1/B2, Galileo E1/E5, GNSS: GPS L1, GLONASS F1, Galileo E1	RTK: GPS L1/L2, GLONASS F1/F2, BeiDou B1/B2, Galileo E1/E5, GNSS: GPS L1, GLONASS F1, Galileo E1
Max Power Consumption	7000 W	5600 W
Hovering Power Consumption	6200 W (with a takeoff weight of 47.5 kg)	4600 W (with a takeoff weight of 39.5 kg)
Hovering Time	15 min (takeoff weight of 27.5 kg with a 18000 mAh battery) 10 min (takeoff weight of 42.6 kg with a 18000 mAh battery)	18 min (takeoff weight of 24.5 kg with a 17500 mAh battery) 10 min (takeoff weight of 39.5 kg with a 17500 mAh battery)
Max Tilt Angle	15°	15°
Max Operating Speed	7 m/s	7 m/s
Max Flying Speed	10 m/s (with a strong GNSS signal)	10 m/s (with a strong GNSS signal)
Max Wind Resistance	8 m/s	8 m/s
Max Service Ceiling Above Sea Level	2000 m	2000 m
Recommended Operating Temperature	0° to 40°C	0° to 40°C
Max Diagonal Wheelbase	1883 mm	1895 mm
Dimensions	2509×2213×732 mm (Arms and propellers unfolded) 1795×1510×732 mm (Arms unfolded and propellers folded) 1100×570×732 mm (Arms and propellers folded)	2520×2212×720 mm (Arms and propellers unfolded) 1800×1510×720 mm (Arms unfolded and propellers folded) 1100×570×720 mm (Arms and propellers folded)
Stator Size	100×15 mm	100×15 mm
Motor Weight	666 g	616 g
KV	75 rpm/v	75 rpm/V
Max Thrust	13.5 kg/rotor	13.5 kg/rotor
Max Power	2400 W/rotor	2400 W/rotor
Diameter × Pitch	33×9 inch	33×9 inch
Weight (Single propeller)	90 g	90 g
Max Working Current (Continuous)	40 A	40 A

Max Working Voltage	58.8 V (14S LiPo)	58.8 V (14S LiPo)
FPV	Horizontal: 98°, Vertical: 78°	Horizontal: 98°, Vertical: 78°
FPV Resolution	1280×960 30fps	1280×960 30fps
FPV Max Brightness	12 lux at 5 m of direct light	12 lux at 5 m of direct light, FOV: 110°
Spray Tank Volume	Rated: 15.1 L, Full: 20 L	Rated: 15.1 L, Full: 16 L
Operating Payload	Rated: 15.1 kg, Full: 20 kg	Rated: 15.1 kg, Full: 16 kg
Nozzle Model	SX11001VS (standard) SX110015VS (optional) XR11002VS (optional)	SX11001VS (standard) SX110015VS (optional)
Quantity	8	8
Max Spray Rate	SX11001VS: 3.6 L/min SX110015VS: 4.8 L/min XR11002VS: 6 L/min	XR11001VS: 3.6 L/min XR110015VS: 4.8 L/min
Droplet Size	SX11001VS: 130-250 μm SX110015VS: 170-265 μm XR11002VS: 190-300 μm (related to the actual working environment, spraying flow, etc.)	SX11001VS: 130-250 μm SX110015VS: 170-265 μm (related to the actual working environment, spraying flow, etc.)
Spray Width	4-7 m (8 nozzles, at a height of 1.5 - 3 m above crops)	4-6.5 m (8 nozzles, at a height of 1.5 - 3 m above crops)
Measurement Range	0.25 - 20 L/min	0.45 - 5 L/min
Error	< ±2%	< ±2%
Measurable Liquid	Conductivity > 50 μS/cm (Liquids such as water or pesticides that contain water)	Conductivity > 50 μS/cm (Liquids such as water or pesticides that contain water)
Radar Model	RD2428R	RD2418R
Radar Operating Frequency	CE (Europe)/(United States): 24.00 GHz-24.25 GHz MIC (Japan)/KCC (Korea): 24.05 GHz-24.25 GHz	SRRC (China)/CE (Europe)/(United States): 24.00 GHz-24.25 GHz MIC (Japan)/KCC (Korea): 24.05 GHz-24.25 GHz
Radar Power Consumption	18 W	15 W
EIRP	MIC / KCC / CE / FCC: < 20 dBm	MIC / KCC / CE / FCC: 20 dBm
Altitude Detection & Terrain Follow	Altitude detection range: 1-30 m Stabilization working range: 1.5-15 m Maximum slope of hill mode: 35°	Altitude detection range: 1-30 m Stabilization working range: 1.5-15 m Maximum slope of hill mode: 35°
Obstacle Avoidance System	Altitude detection range: 1.5-30 m FOV: Horizontal: 360°, Vertical: ±15° Condition: Relative altitude higher than 1.5 m, operation speed lower than 7 m/s Safe distance: 2.5m (distance between the propeller tip and the obstacle after the aircraft stops) Obstacle-avoiding direction: Omnidirectional obstacle sensing operates horizontally and covers 360°	Altitude detection range: 1.5-30 m FOV: Horizontal: ±50°, Vertical: 0° to 10° Condition: Relative altitude higher than 1.5 m, operation speed lower than 7 m/s Safe distance: 2.5m (distance between the propeller tip and the obstacle after the aircraft stops) Obstacle-avoiding direction: Forward and backward
IP Rating	IP67	IP67
Intelligent Flight Battery Model	AB3-18000mAh-51.8V	AB2-17500mAh-51.8V
Remote Controller Model	RM500-AG	GL300N
Operating Frequency	2.4000-2.4835 GHz 5.725-5.850 GHz	2.4000 GHz-2.4835 GHz 5.725 GHz- 5.850 GHz (not available in Japan, Russia, Ukraine)

Effective Transmission Distance (Unobstructed, free of interference)	SRRC/MIC/KCC/CE: 3 km NCC/FCC: 5km (Operation height: 2.5 m)	SRRC/MIC/KCC/CE: 3 km NCC/FCC: 5km
Transmission Power (EIRP)	2.4 GHz SRRC/CE/MIC/KCC: 18.5 dBm, FCC: 25.5 dBm; 5.8 GHz SRRC/FCC: 25.5 dBm	2.4 GHz: SRRC / CE / MIC / KCC: < 20 dBm FCC / NCC: < 26 dBm 5.8 GHz: SRRC / NCC / FCC: < 26 dBm KCC / CE: < 14 dBm
Display Device	5.5-inch screen 1920×1090 pixels Brightness: 1000 cd/m² Android System	5.5-inch screen 1920×1090 pixels Brightness: 1000 cd/m² Android System
Random Access Memory, RAM	4GB LPDDR4	4G
Storage Capacity	ROM 32GB + microSD, supports microSD cards with a capacity of up to 128 GB and R/W speed up to UHS-I Speed Grade 3	16G
Remote Controller Power Consumption	18 W	16 W (Typical value)
Remote Controller Operation Temperature Range	-10° to 40°C	-10° to 40°C
Remote Controller Storage Temperature Range	- < 1 month: -30° to 60°C 1-3 months: -30° to 45°C 3-6 months: -30° to 35°C > 6 months: -30° to 25°C (Built-in battery power 40% to 60%)	Less than 3 months: -20° to 45°C (68° to 113°F) More than 3 months: 22° to 28°C (70° to 82°F)
Remote Controller Charging Temperature	5° to 40°C	5° to 40°C
Supported Aircraft	T20, T16	T16
Remote Controller Intelligent Battery Model	WB37-4920mAh-7.6V	WB37-4920mAh-7.6V
Remote Controller Intelligent Battery Type	2S LiPo	2S LiPo
Remote Controller Intelligent Battery Capacity	4920 mAh	4920 mAh
Remote Controller Intelligent Battery Voltage	7.6 V	7.6 V
Remote Controller Intelligent Battery Energy	37.39 Wh	37.39 Wh
Remote Controller Intelligent Battery Charging Temperature	5° to 40°C	5° to 40°C
Remote Controller Intelligent Battery Working Time	2 hours	---

Note: The above material number, material name, disassembly and assembly operations, and troubleshooting will be updated continuously, and they are for reference only.

大疆农业售后外发文档 (代理商专用)